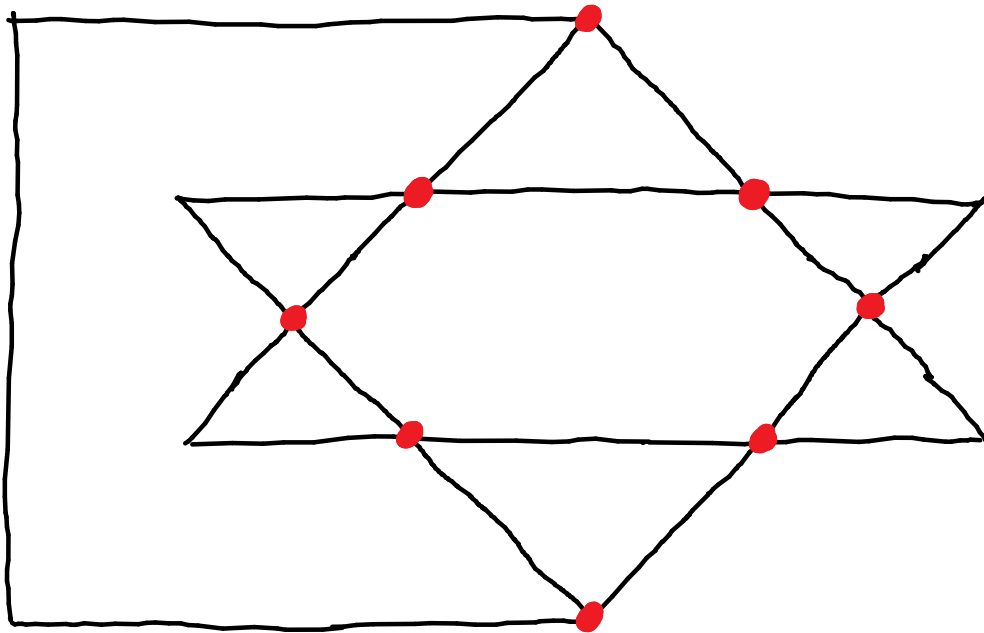


EBN111/122

Tutorial 2 – Memo 29 July 2017

1

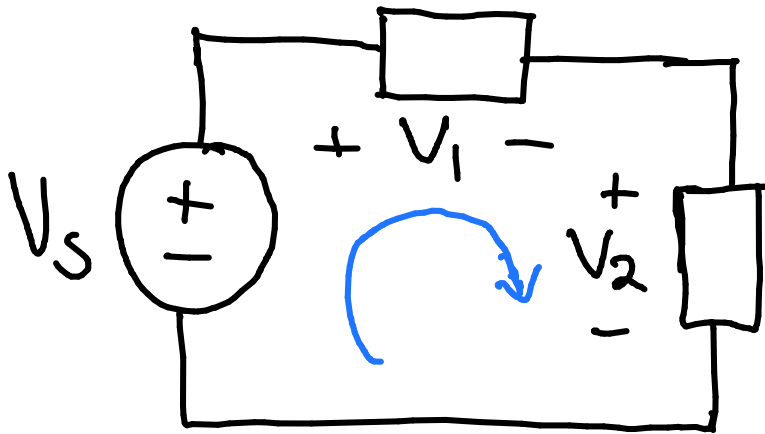
2.6 In the network shown below, determine the number of branches, nodes and loops.



$$\begin{aligned} \text{nodes} &= 8 \\ \text{loops} &= 8 \end{aligned}$$

$$\begin{aligned} b &= n + l - 1 \\ &= 8 + 8 - 1 \\ &= 15 \end{aligned}$$

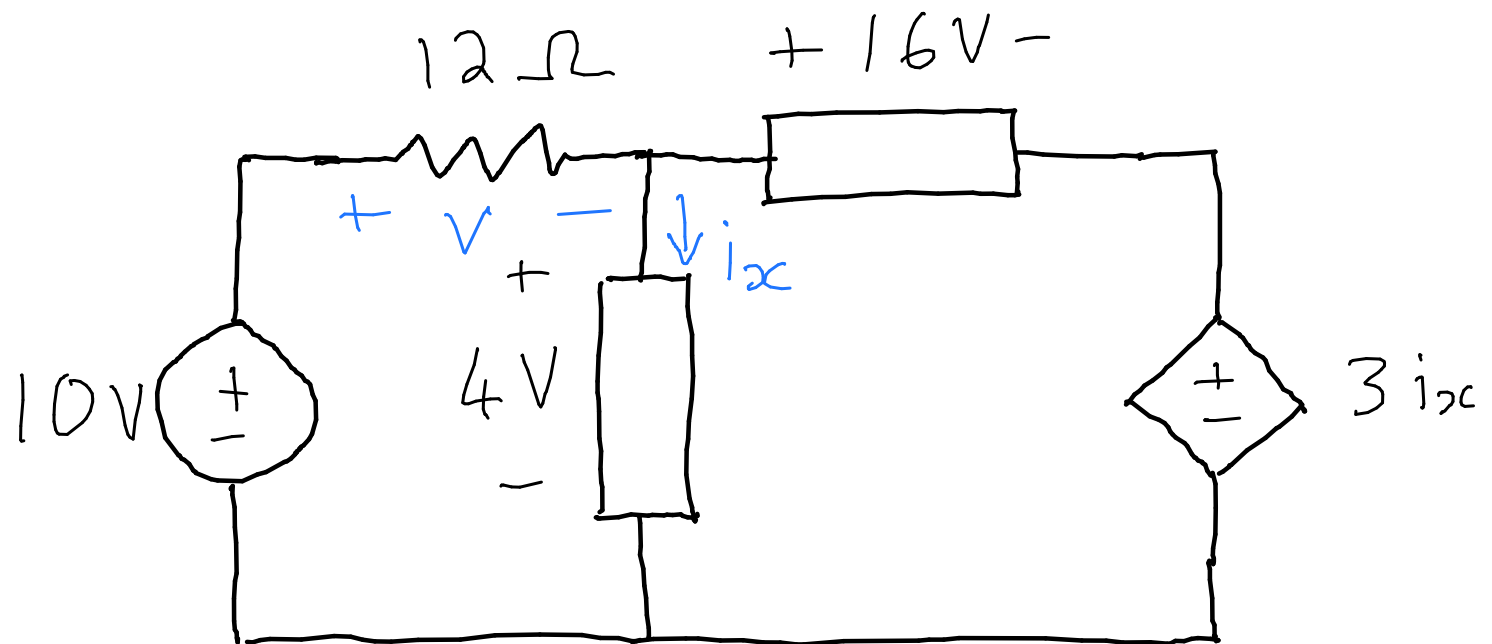
KVL: The sum of voltages in a loop is equal to 0.

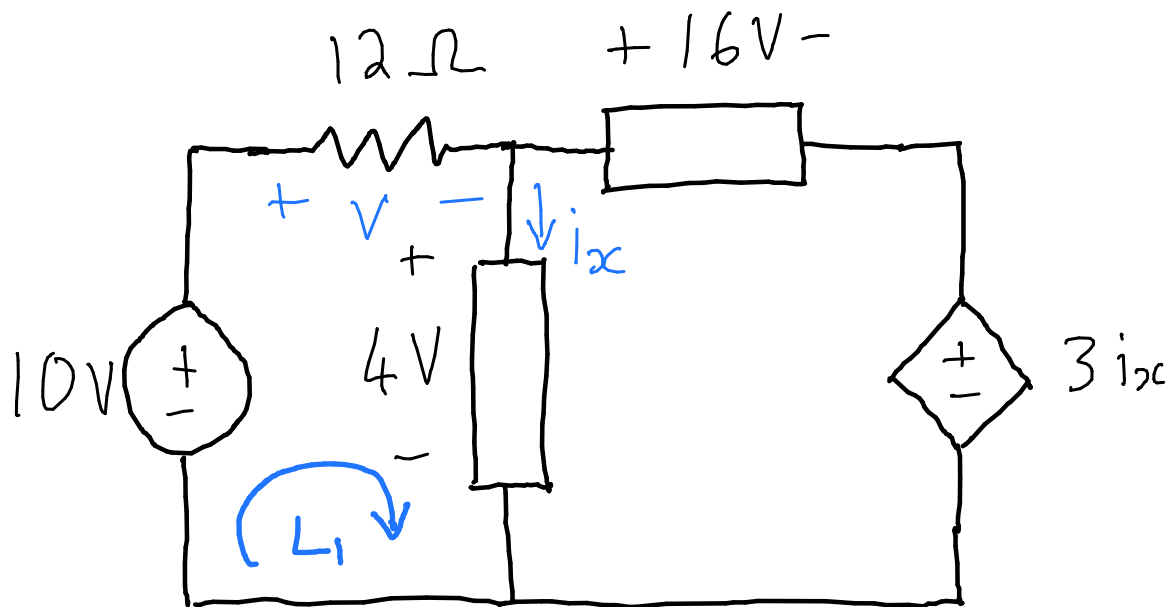


$$\sum V_n = 0$$

$$-V_s + V_1 + V_2 = 0$$

2.15 Calculate v and i_x in the circuit below.



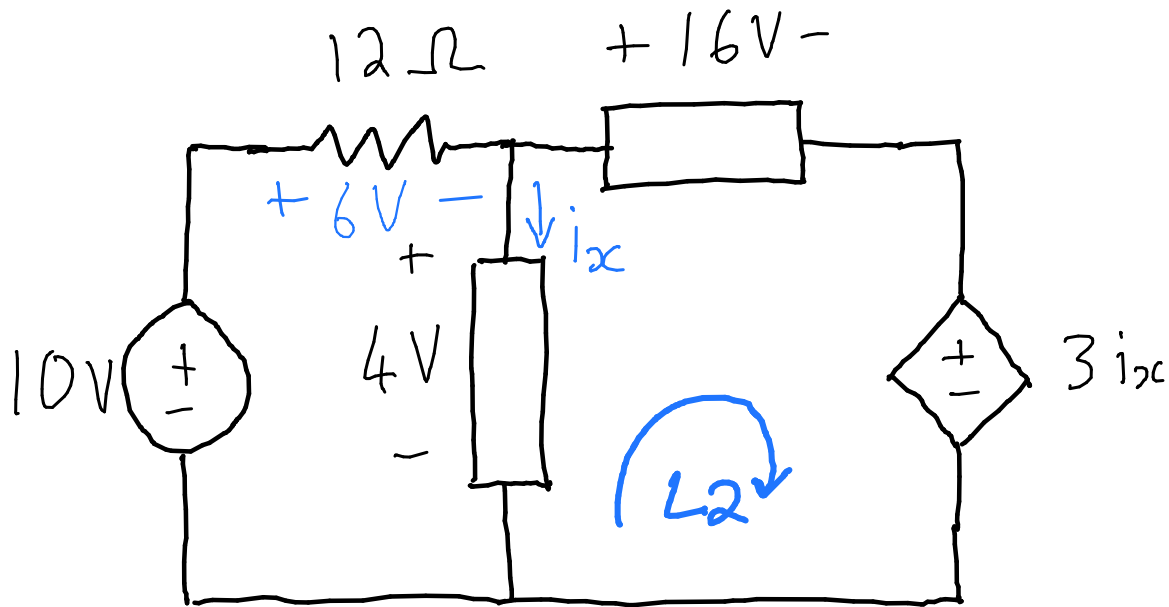


KVL L_1 :

$$-10 + V + 4 = 0$$

$$V = 10 - 4$$

$$V = 6V$$



KVL L_2 :

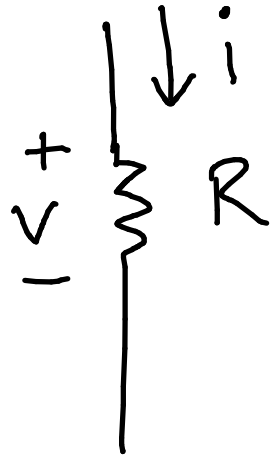
$$-4 + 16 + 3i_x = 0$$

$$3i_x = 4 - 16$$

$$i_x = -12/3$$

$$i_x = -4 \text{ A}$$

Ohm's law:



$$V = iR$$

$$i = \frac{V}{R}$$

$$i = VG$$

$$V = \frac{i}{G}$$

$$G = \frac{1}{R}$$

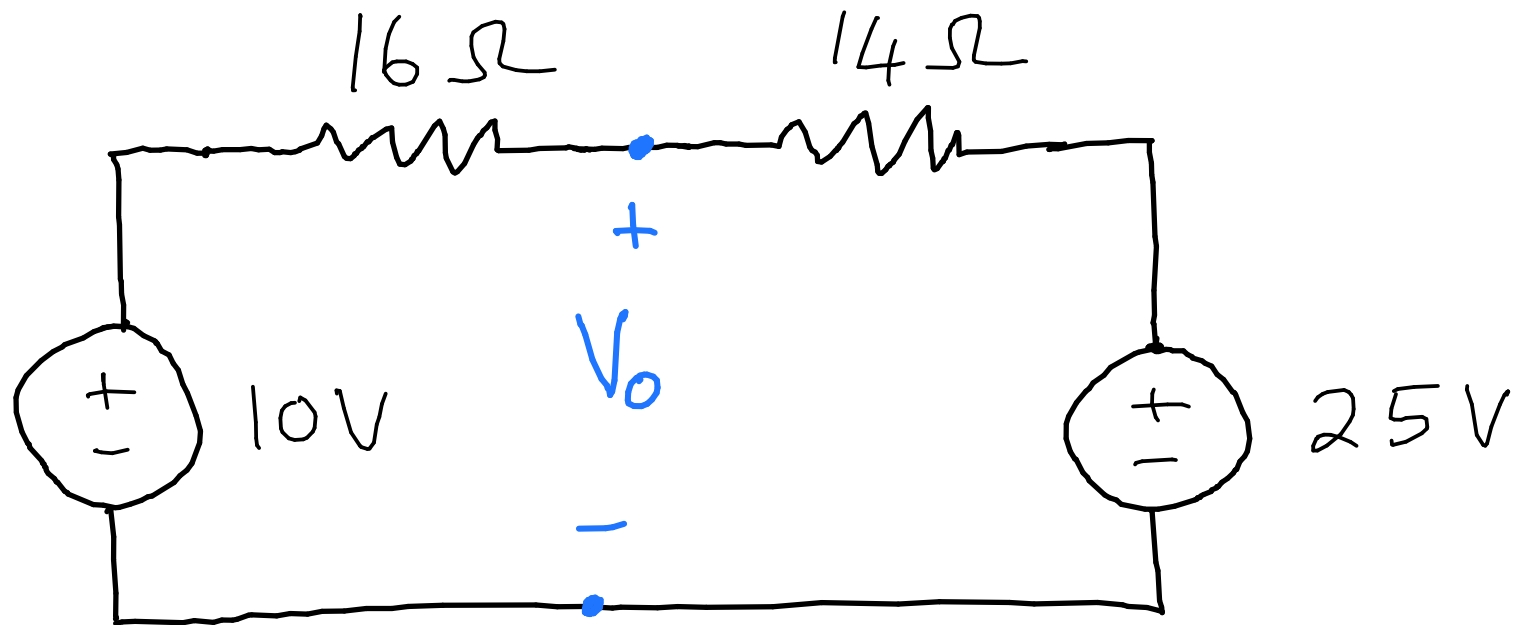
$$V[V]$$

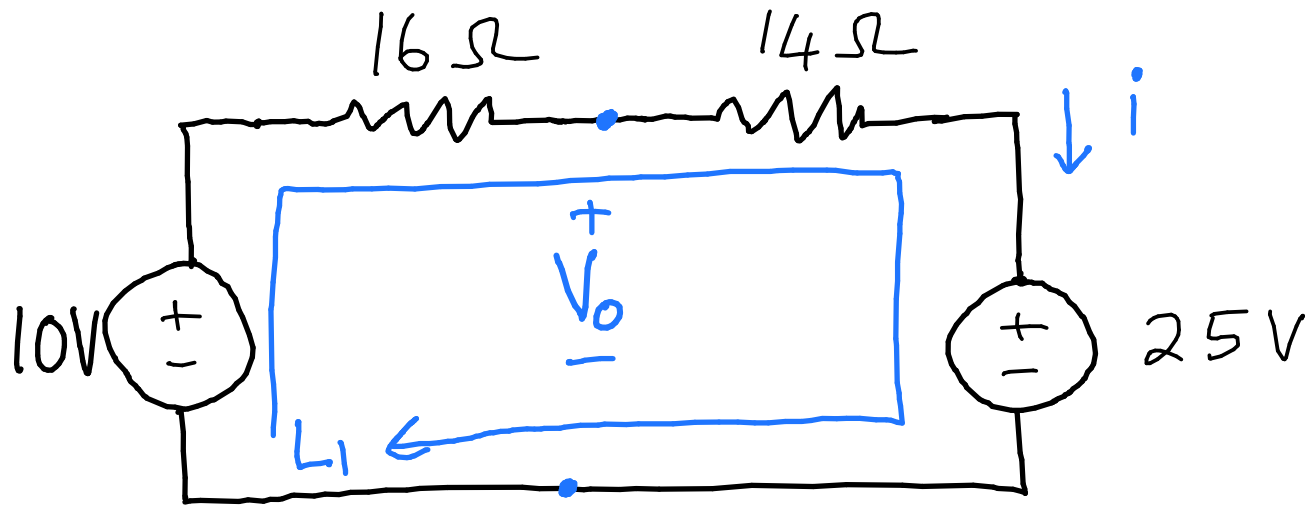
$$i[A]$$

$$R[\Omega]$$

$$G[S]$$

2.16 Determine V_o in the circuit below.





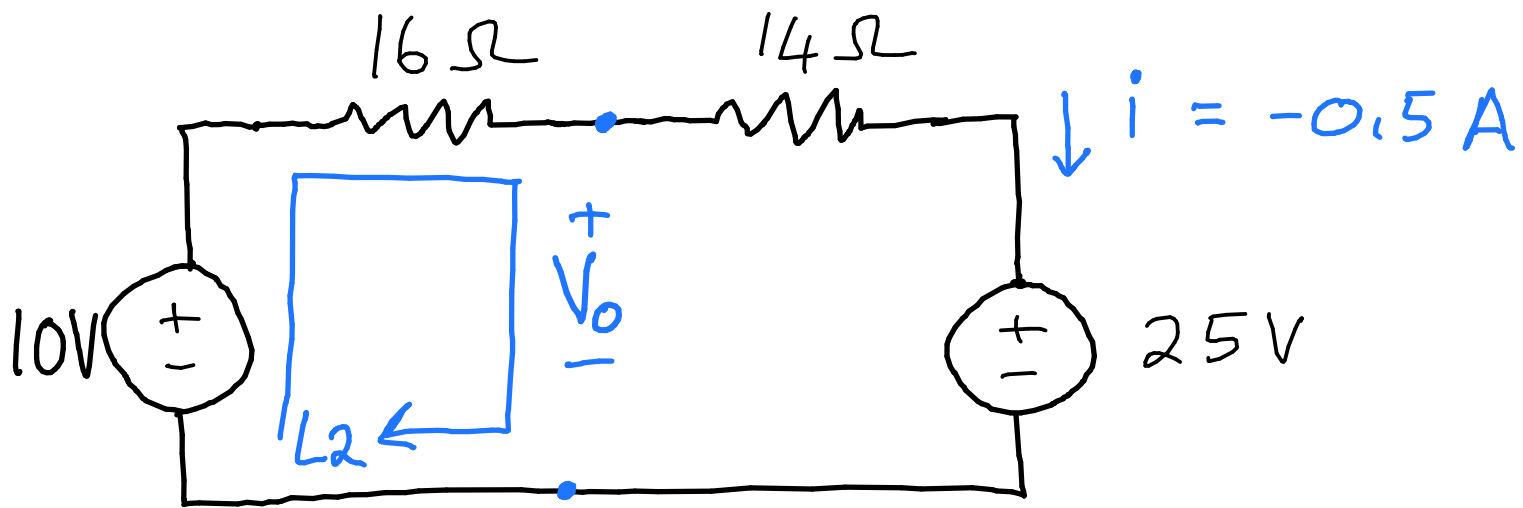
KVL in loop 1:

$$-10 + 16i + 14i + 25 = 0$$

$$16i + 14i = 10 - 25$$

$$30i = -15$$

$$i = -0.5 \text{ A}$$



KVL in loop 2

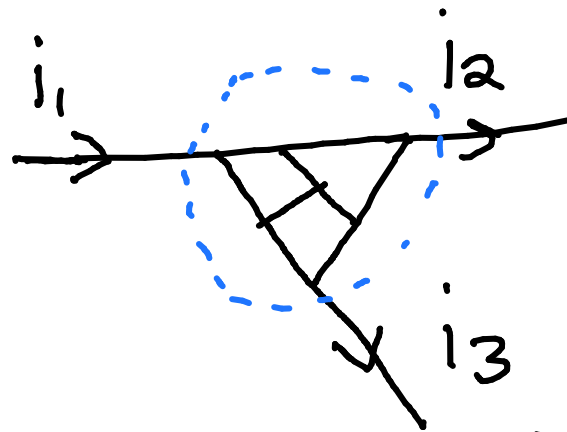
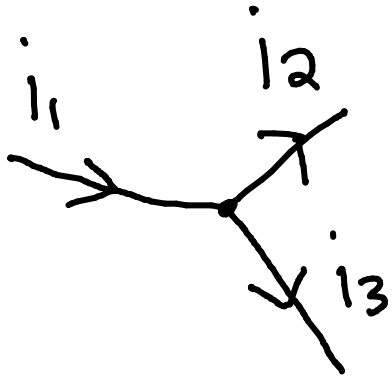
$$-10 + 16i + V_0 = 0$$

$$-10 + 16(-0.5) + V_0 = 0$$

$$V_0 = 10 + 8$$

$$V_0 = 18 \text{ V}$$

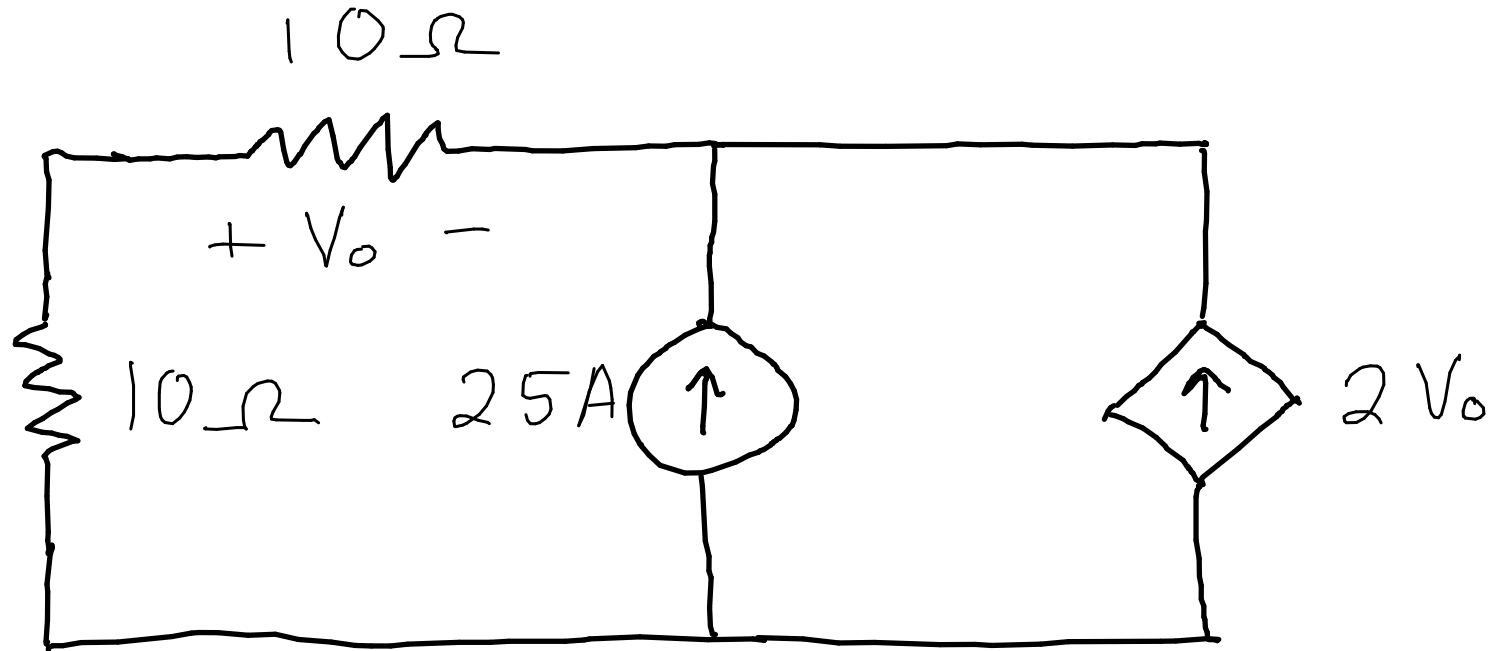
KCL: The sum of currents entering a node (or closed boundary) is zero.

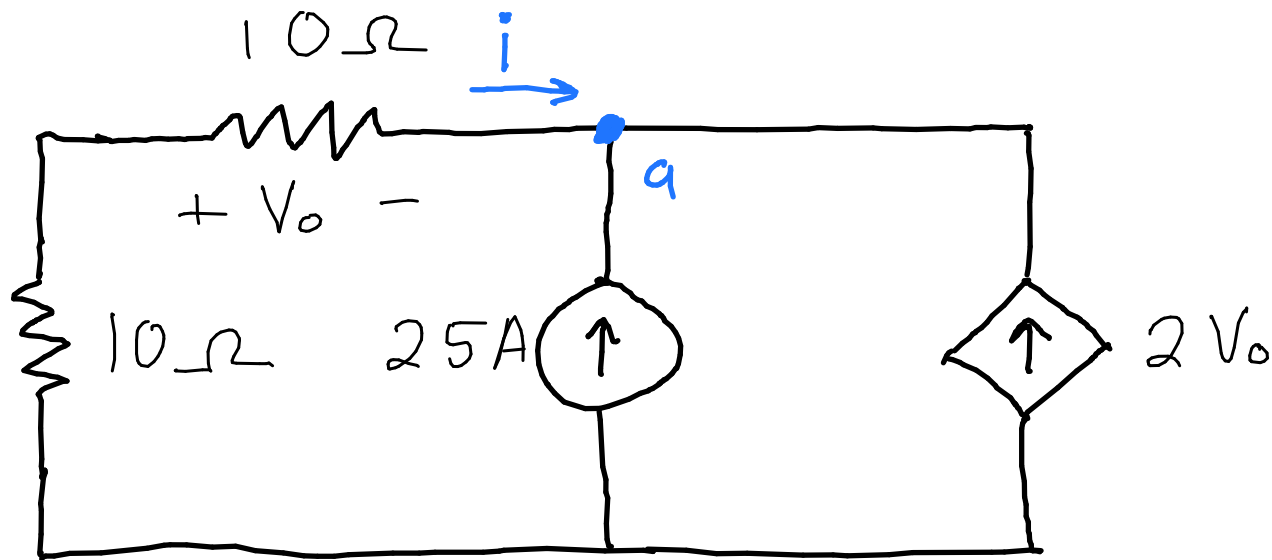


$$i_1 + (-i_2) + (-i_3) = 0$$

$$i_1 = i_2 + i_3$$

2.22 Find V_o in the circuit below, and the power absorbed by the dependent source.





Ohm's
Law:

$$i = \frac{V_o}{10}$$

KCL at node a :

$$i + 25 + 2V_o = 0$$

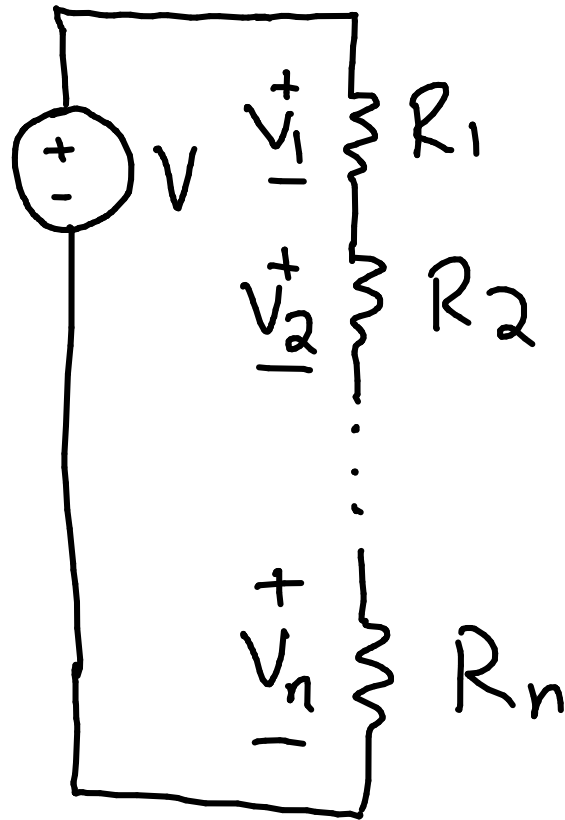
$$\frac{V_o}{10} + 2V_o = -25$$

$$V_o + 20V_o = -250$$

$$21V_o = -250$$

$$V_o = -11,9048V$$

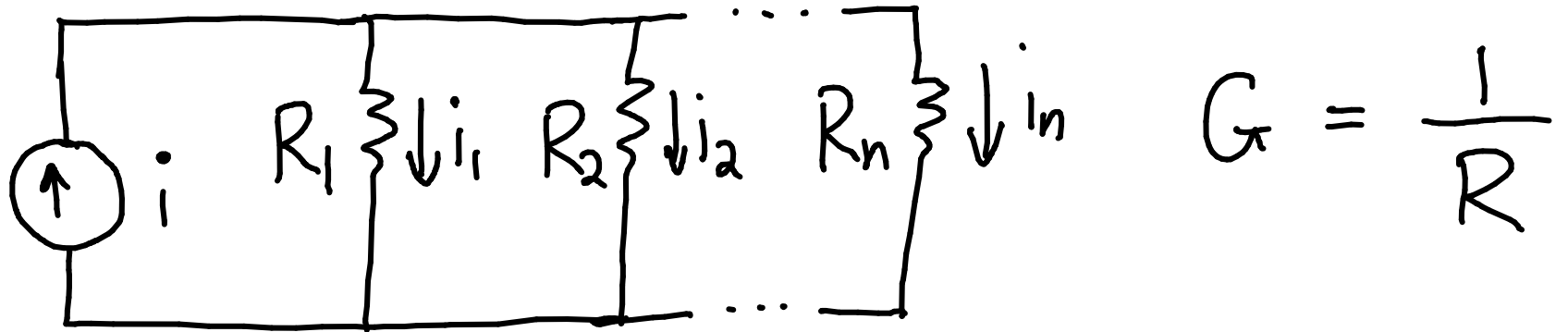
Series Resistors & Voltage Division



$$R_{eq} = R_1 + R_2 + \dots + R_n$$

$$V_n = \frac{V R_n}{R_1 + R_2 + \dots + R_n}$$

Parallel resistors & current division

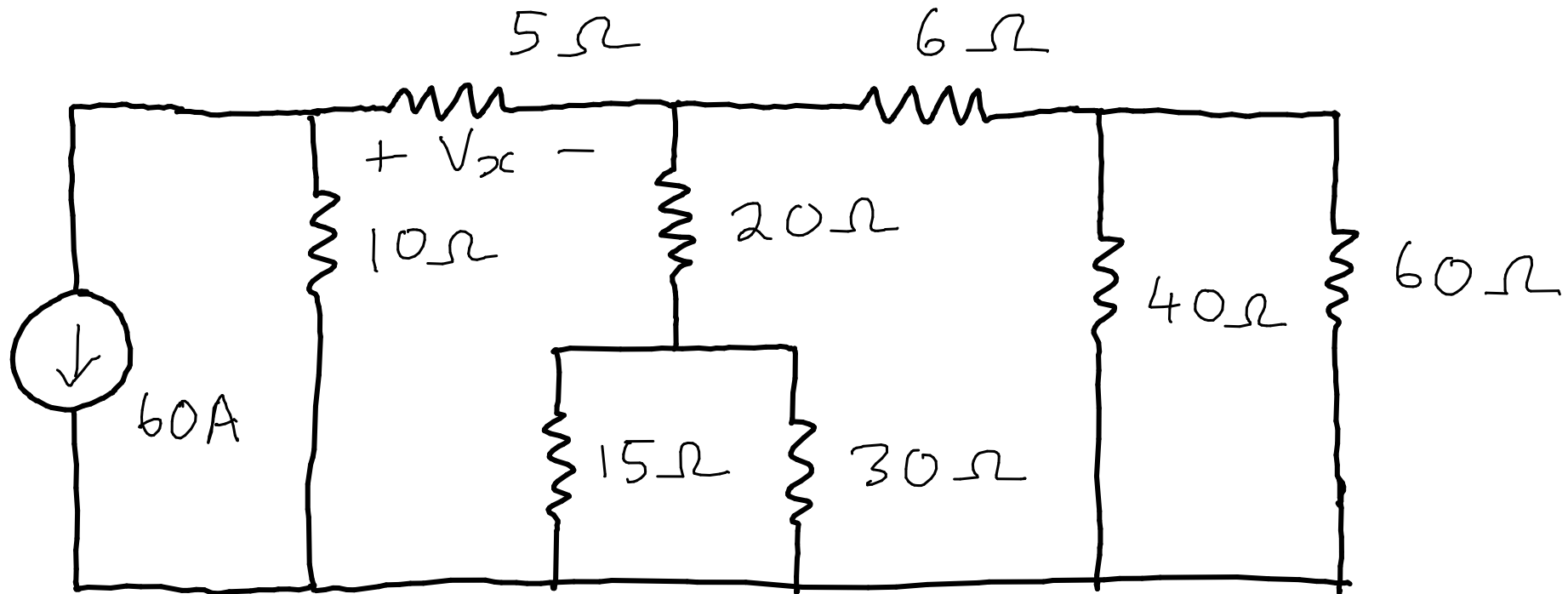


$$G_{eq} = G_1 + G_2 + \dots + G_n$$

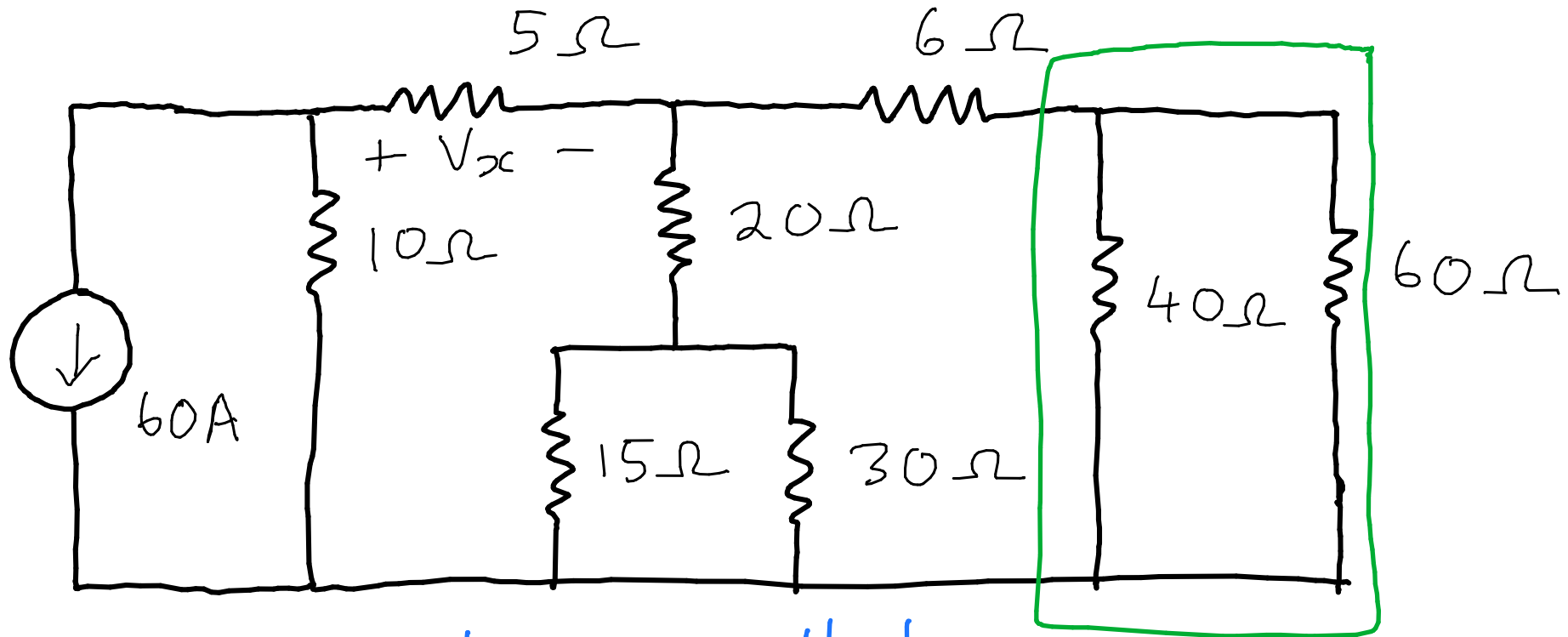
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

$$i_n = \frac{i G_n}{G_1 + G_2 + \dots + G_n} = \frac{i}{R_n} \left(\frac{1}{R_1} + \dots + \frac{1}{R_n} \right)^{-1}$$

2.23 In the circuit shown, determine V_x and the power absorbed by the $60\ \Omega$ resistor.



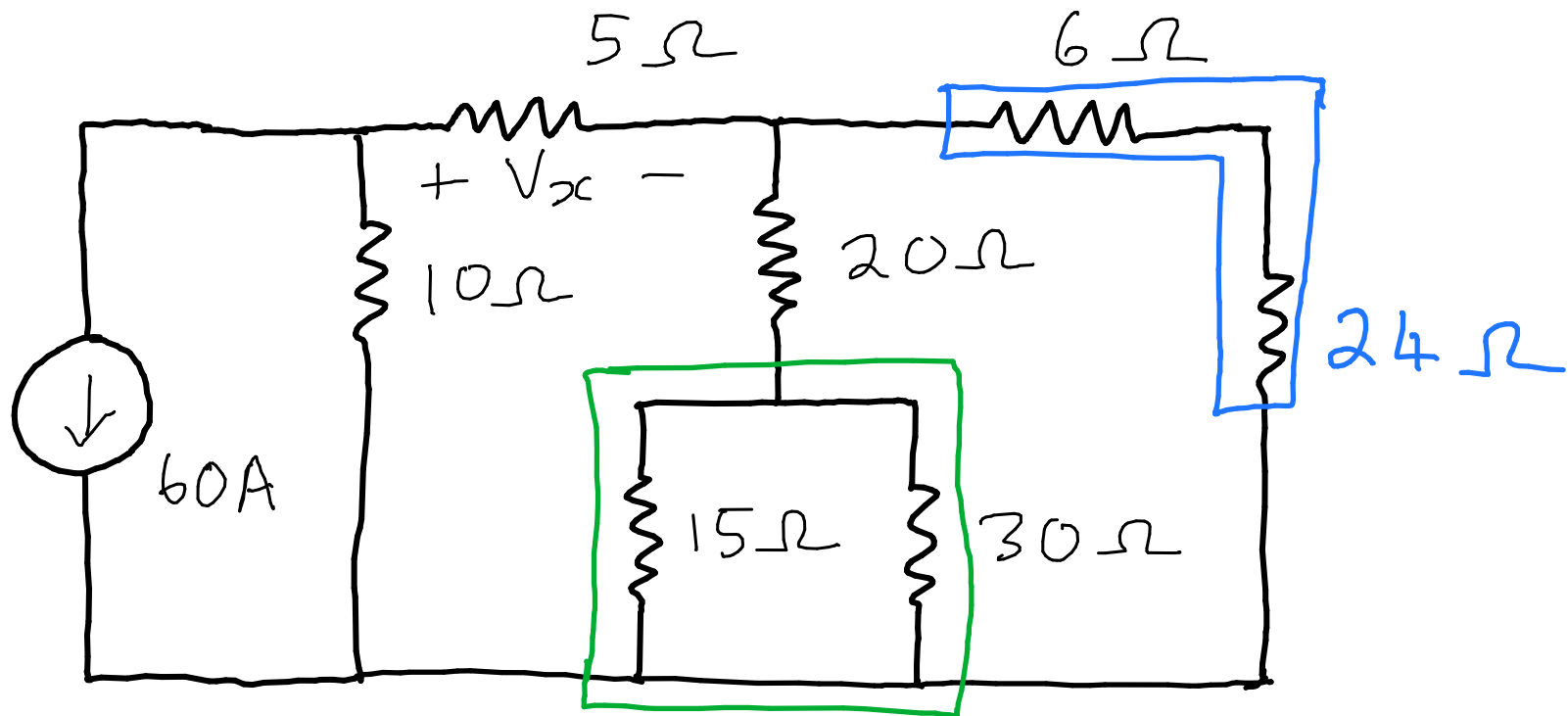
Simplify circuit to find V_x



40Ω is in parallel with 60Ω

$$40 \parallel 60 = \left(\frac{1}{40} + \frac{1}{60} \right)^{-1}$$

$$= 24 \Omega$$

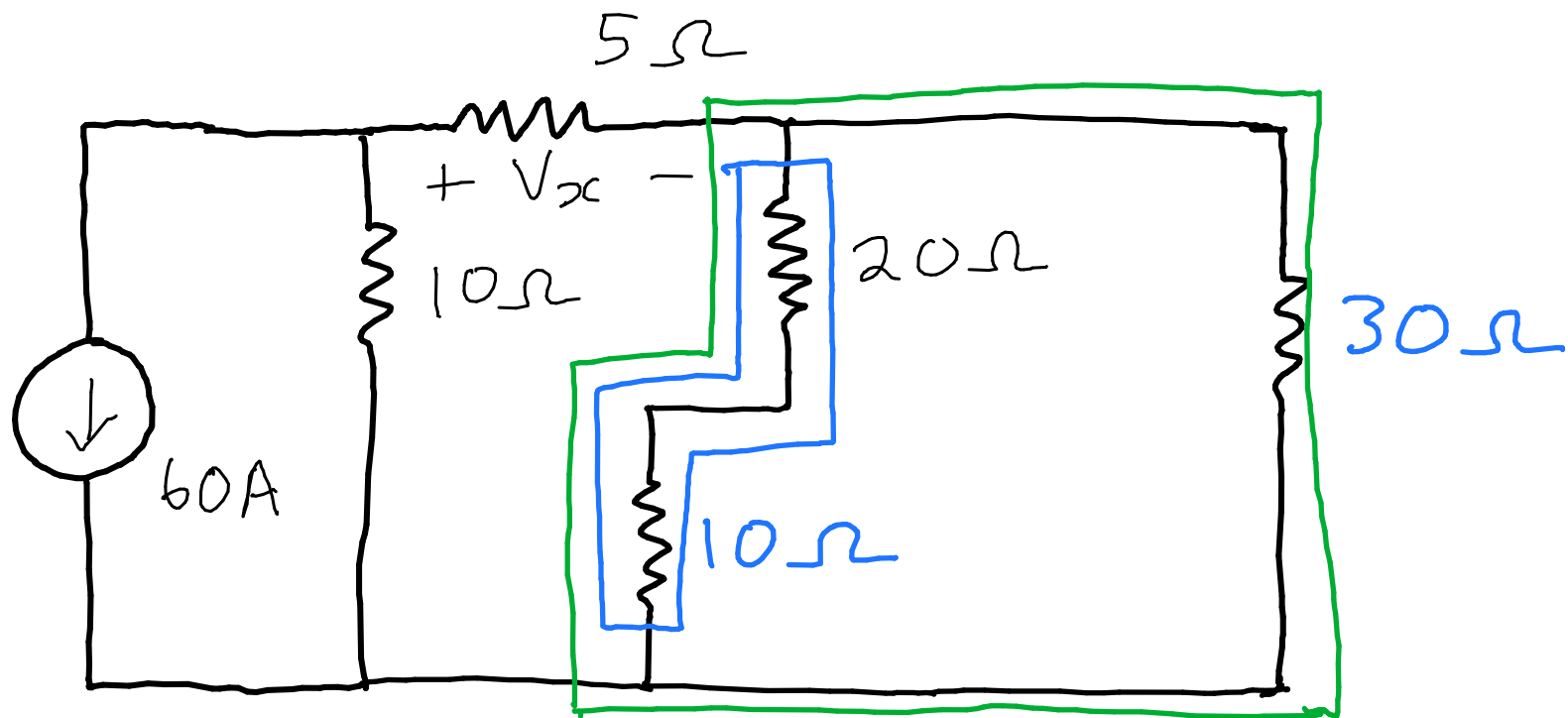


6Ω is in series with 24Ω

$$6 + 24 = 30\Omega$$

$$15 \parallel 30 = \left(\frac{1}{15} + \frac{1}{30} \right)^{-1}$$

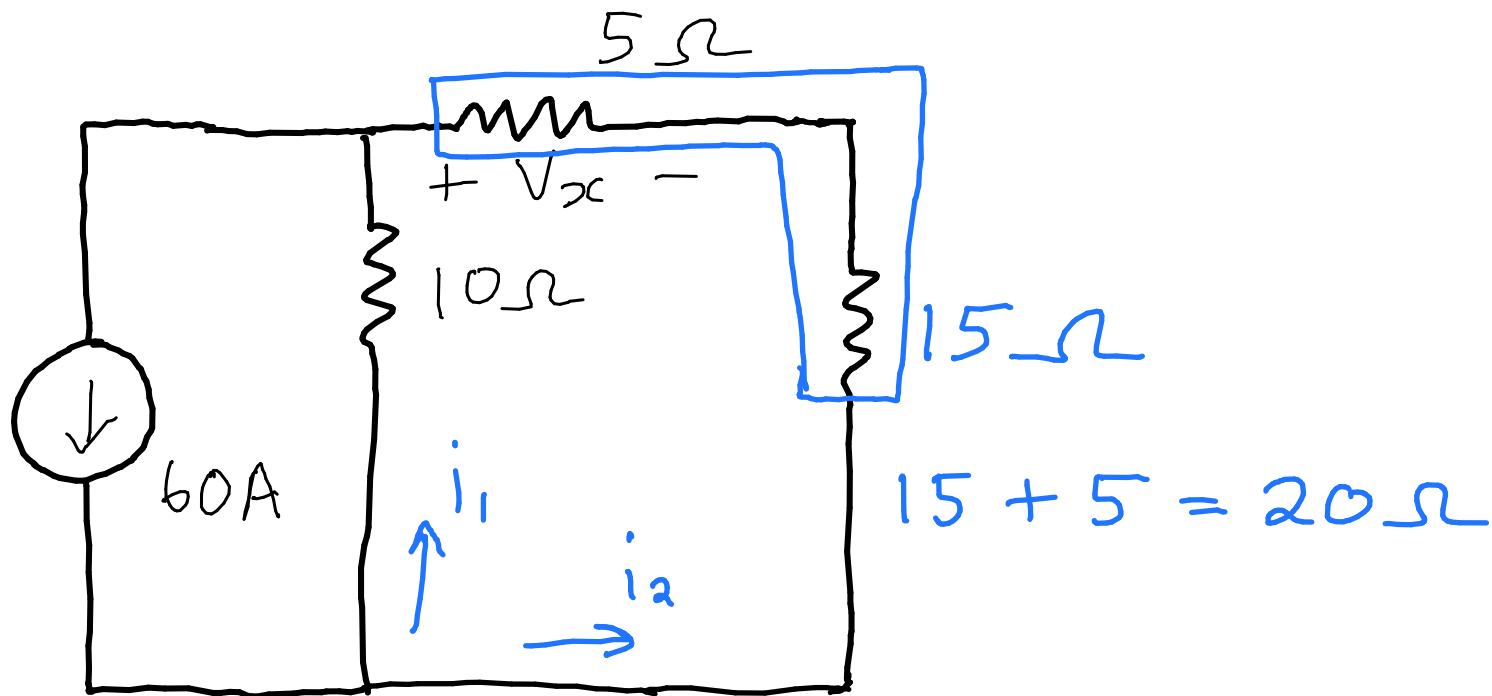
$$= 10\Omega$$



$$10 + 20 = 30\Omega$$

$$30 || 30 = \left(\frac{1}{30} + \frac{1}{30} \right)^{-1}$$

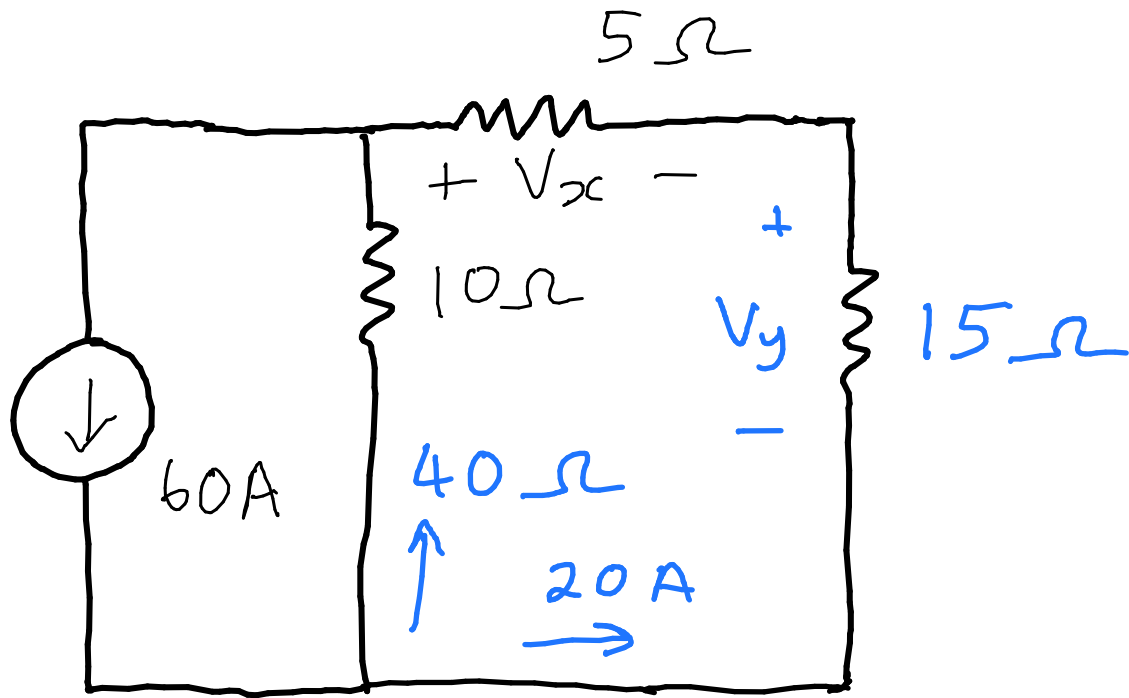
$$= 15\Omega$$



$$i_2 = \frac{i G_2}{G_1 + G_2}$$

$$= \frac{60}{20} \times \left(\frac{1}{20} + \frac{1}{10} \right)^{-1}$$

$$= 20\text{A}$$

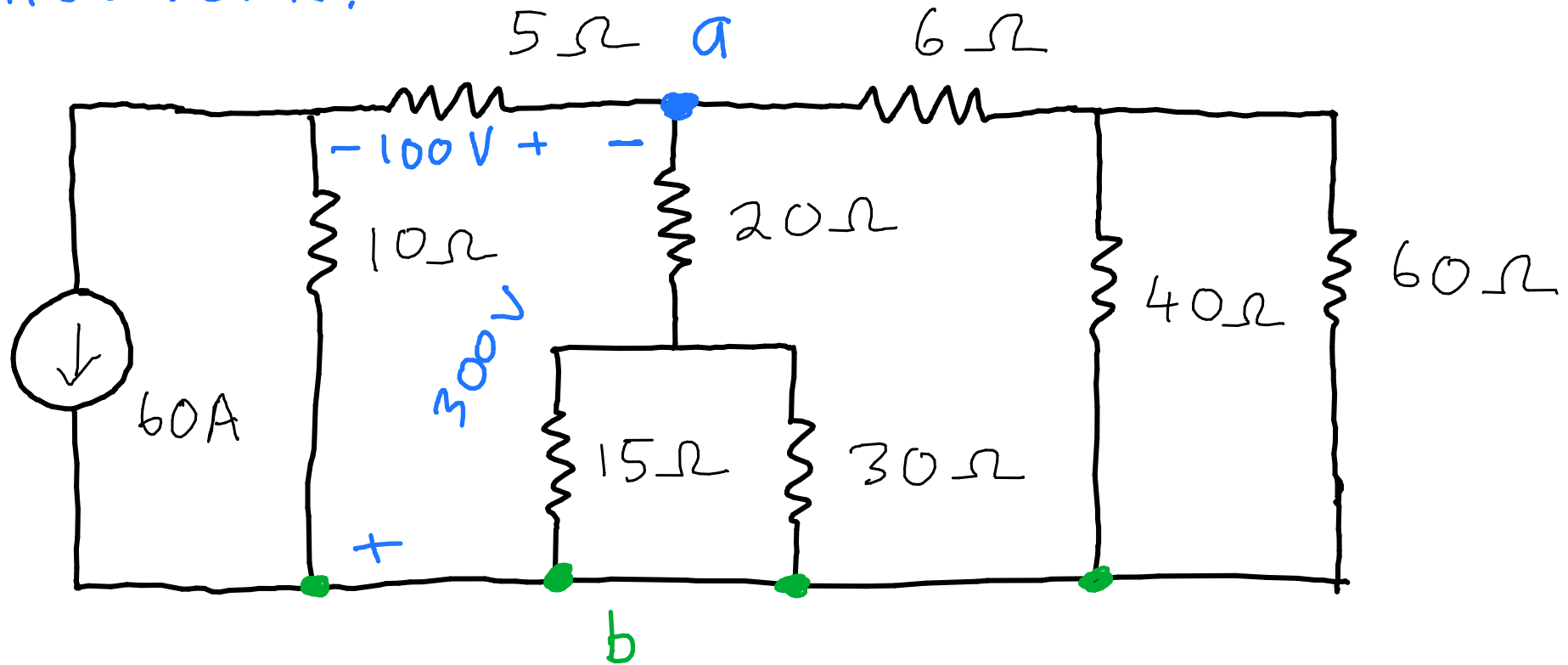


Using Ohm's law & Passive sign Convention.

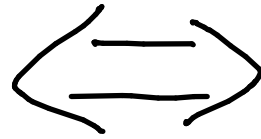
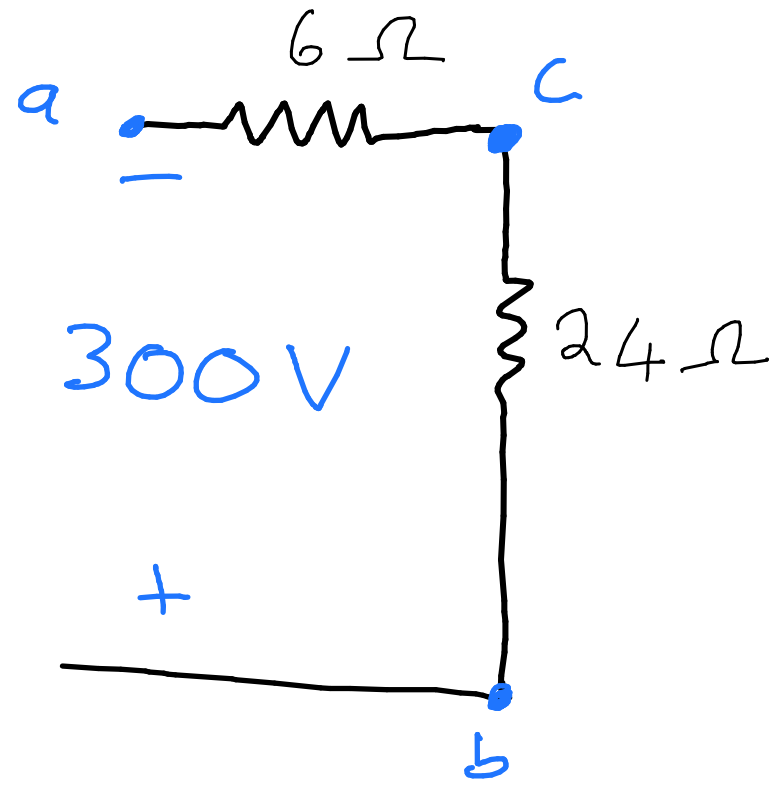
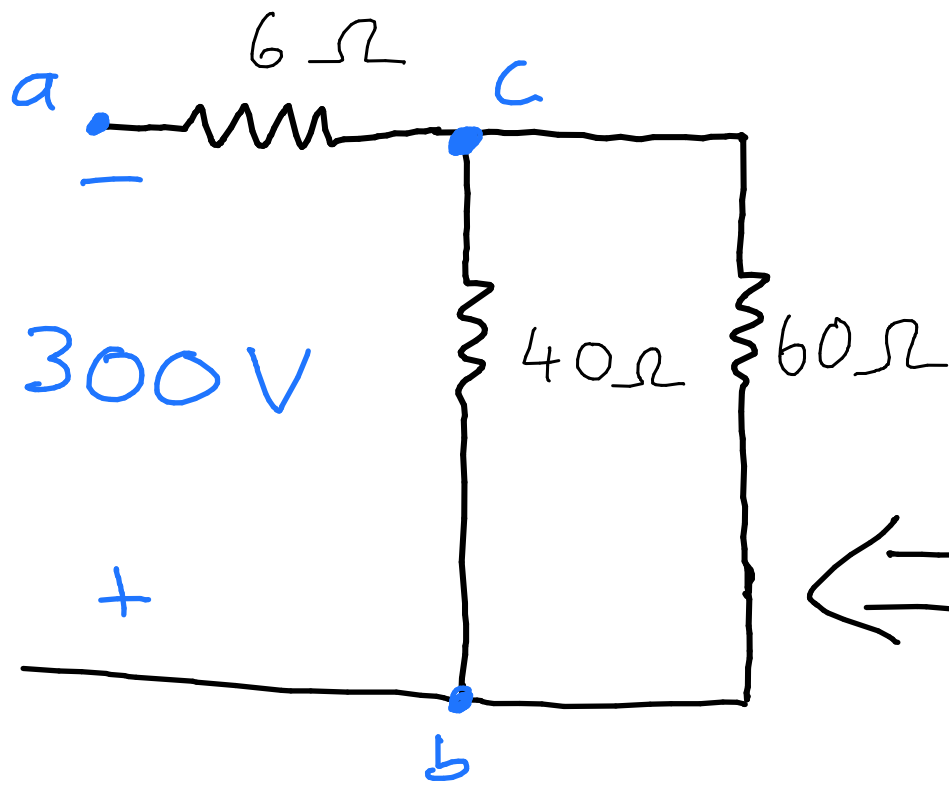
$$\begin{aligned}
 V_x &= -i_2 R_x \\
 &= -20 \times 5 \\
 &= -100 \text{ V}
 \end{aligned}$$

$$\begin{aligned}
 V_y &= -i_2 R_y \\
 &= -20 \times 15 \\
 &= -300 \text{ V}
 \end{aligned}$$

Expand 15Ω resistor into original network.



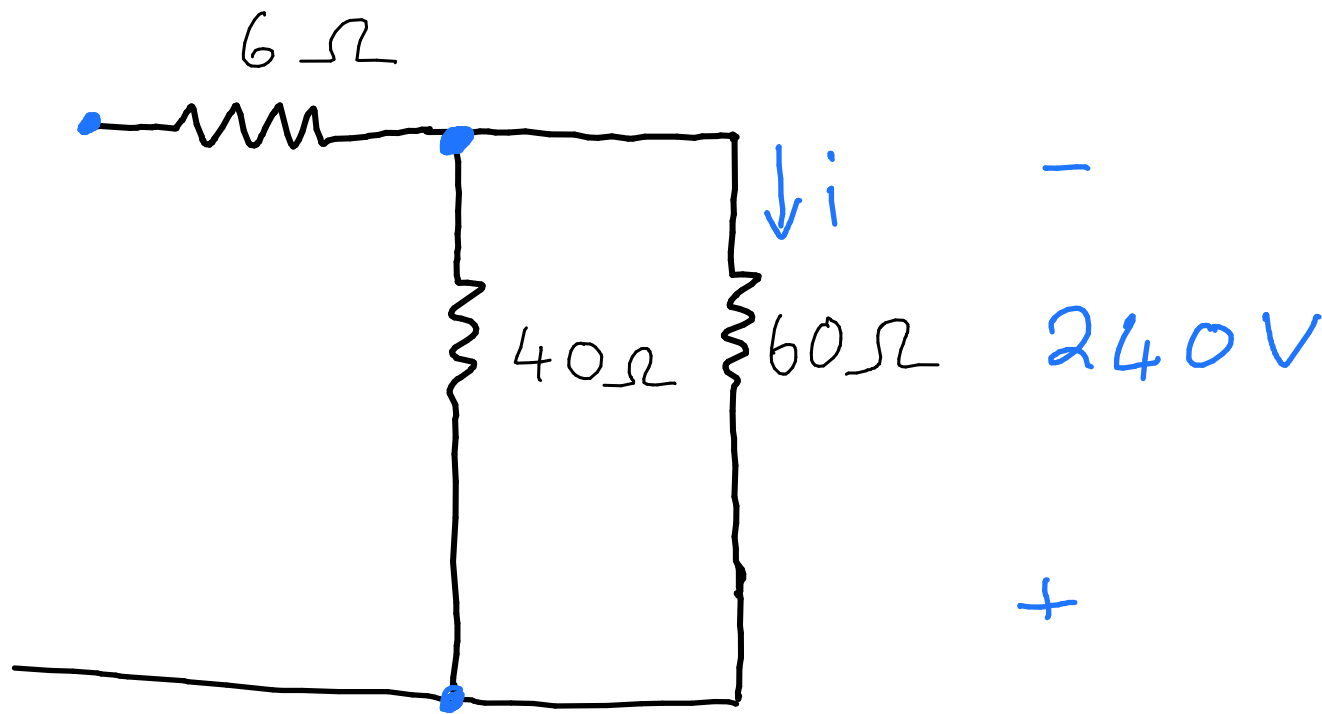
$$V_{ab} = -300V$$



Use voltage division to find V_{cb} .

$$V_{cb} = \frac{V_{ab} \times R_1}{R_1 + R_2}$$

$$V_{cb} = \frac{-300 \times 24}{24 + 6} = -240V$$



Using Ohm's law

$$i = \frac{V}{R}$$

$$i = \frac{-240}{60} = -4\text{ A}$$

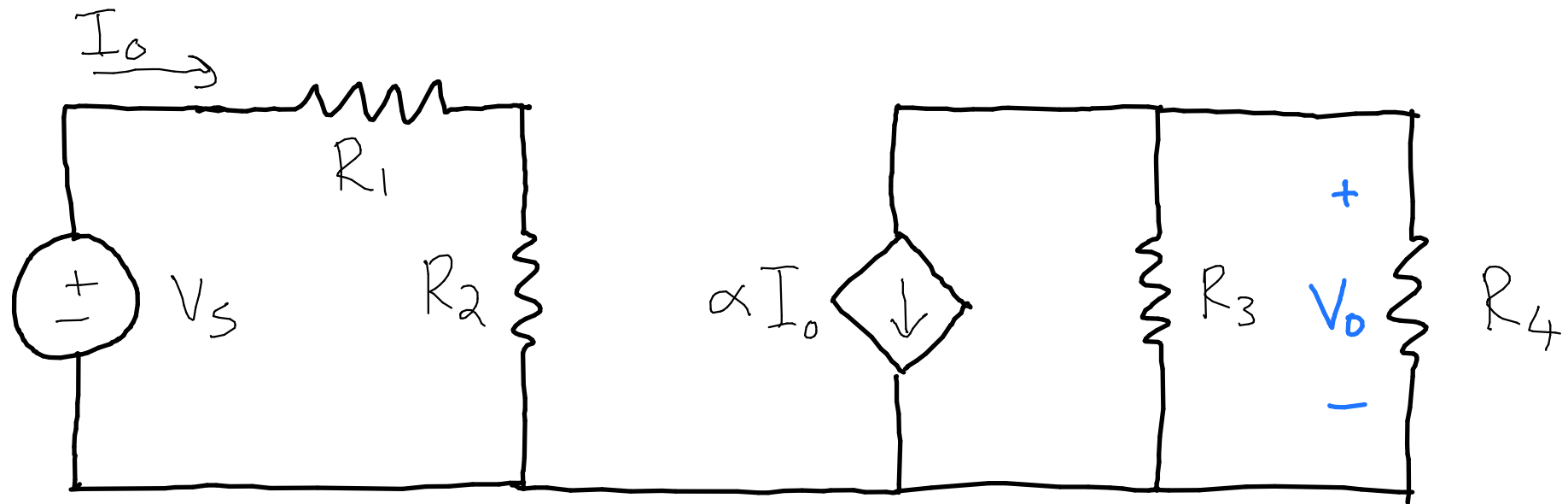
power law

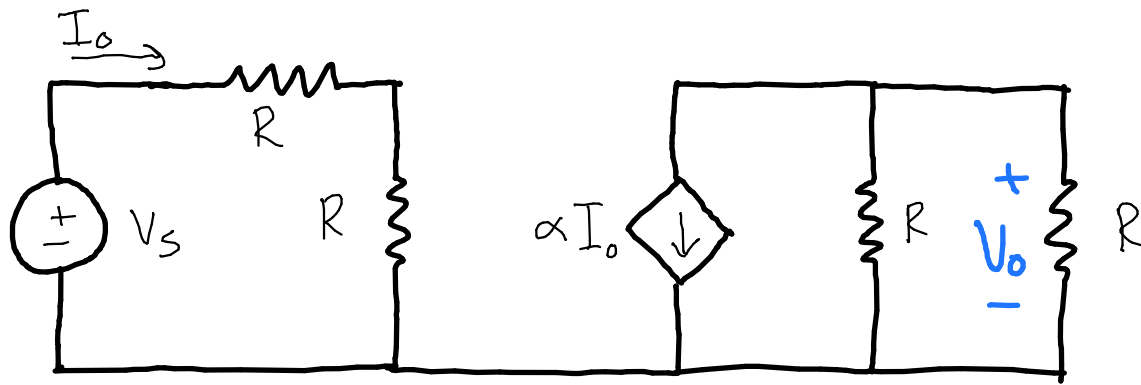
$$P = V \cdot i$$

$$= (-240)(-4)$$

$$= 960\text{ W}$$

2.24 Find $\frac{V_o}{V_s}$ in terms of α , R_1 , R_2 , R_3 , R_4 , if $R_1 = R_2 = R_3 = R_4$



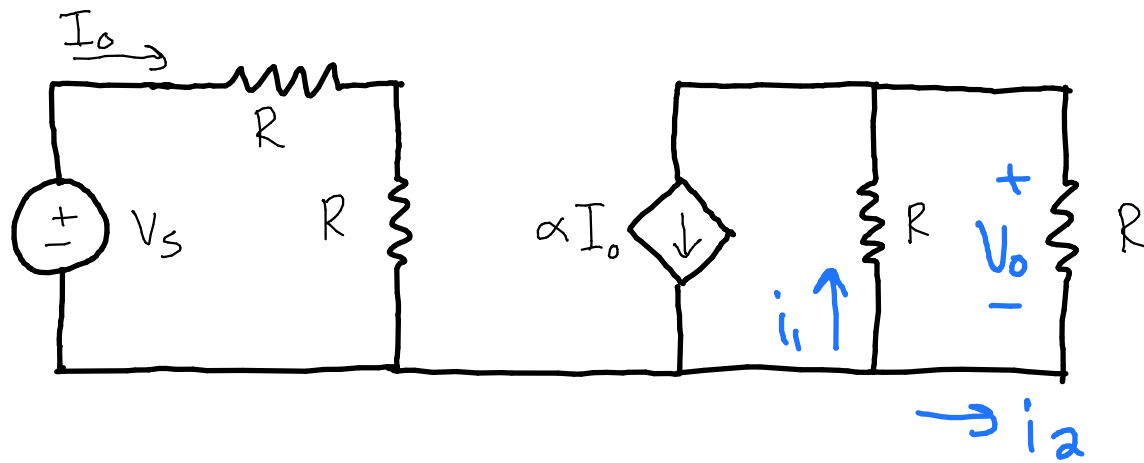


Using ohm's law

$$i = \frac{V}{R}$$

$$I_o = \frac{V_s}{2R}$$

$$\alpha I_o = \frac{\alpha V_s}{2R}$$



$$\alpha I_o = \frac{\alpha V_s}{2R}$$

Current division

$$i_2 = \frac{i_1 G_2}{G_1 + G_2}$$

$$i_2 = \frac{\alpha I_o \times 1/R}{1/R + 1/R}$$

$$= \frac{\alpha I_o}{2}$$

Ohm's law

$$V_o = -i_2 R$$

$$V_o = \frac{-\alpha I_o R}{2}$$

$$\frac{-\alpha I_o R}{2} = V_o$$

$$\alpha I_o = \frac{-2V_o}{R} = \frac{\alpha V_s}{2R}$$

$$\frac{V_o}{V_s} = \frac{\alpha}{2R} \times \frac{-R}{2}$$

$$\frac{V_o}{V_s} = \frac{-\alpha}{4}$$

What value of α will produce

$$\left| \frac{V_o}{V_s} \right| = 10$$

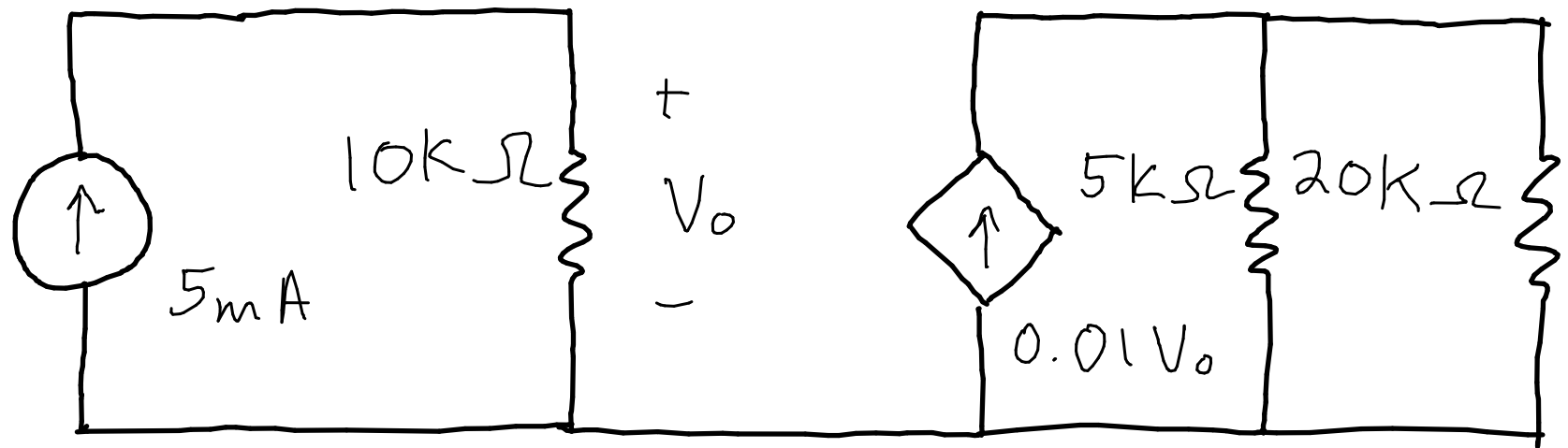
$$\left| \frac{V_o}{V_s} \right| = 10$$

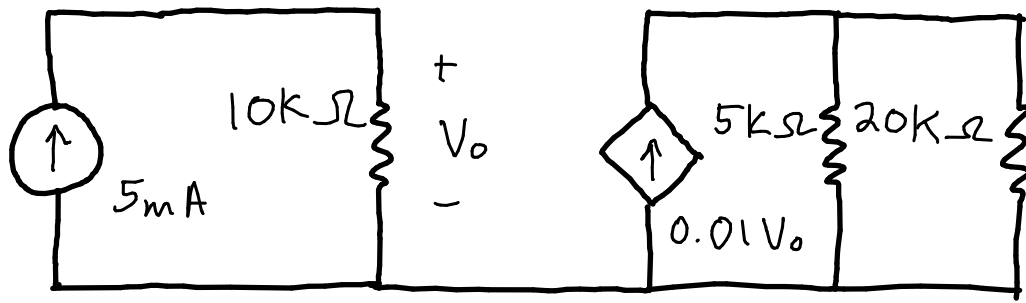
$$\left| \frac{-\alpha}{4} \right| = 10$$

$$\frac{\alpha}{4} = 10$$

$$\alpha = 40$$

2.25 Find the current, voltage and power associated with the $20\text{ k}\Omega$ resistor.



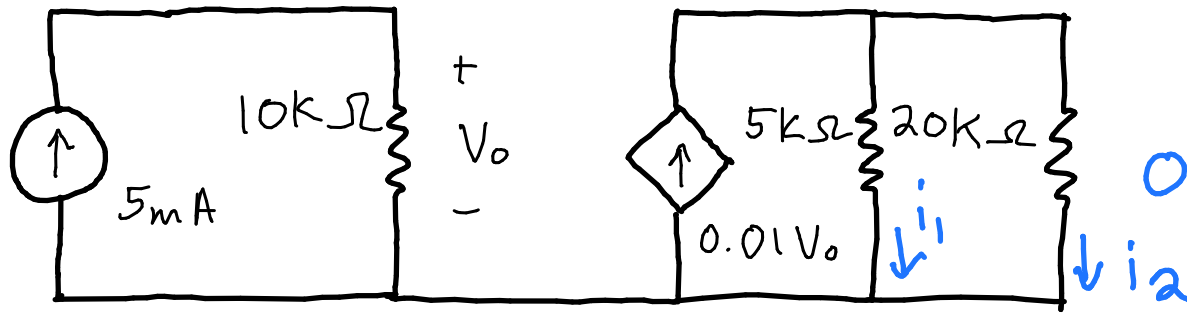


Ohm's law

$$V_o = i \cdot R$$

$$= (5 \times 10^{-3}) \times (10 \times 10^3)$$

$$= 50 \text{ V}$$



$$V_o = 50V$$

$$0.01V_o = 0.5V$$

Current division

$$i_2 = \frac{i G_2}{G_1 + G_2}$$

$$= \frac{0.5}{20 \times 10^3} \times \left(\frac{1}{20 \times 10^3} + \frac{1}{5 \times 10^3} \right)^{-1}$$

$$= 0.1 A$$

$$i_{20k} = 0.1 \text{ A}$$

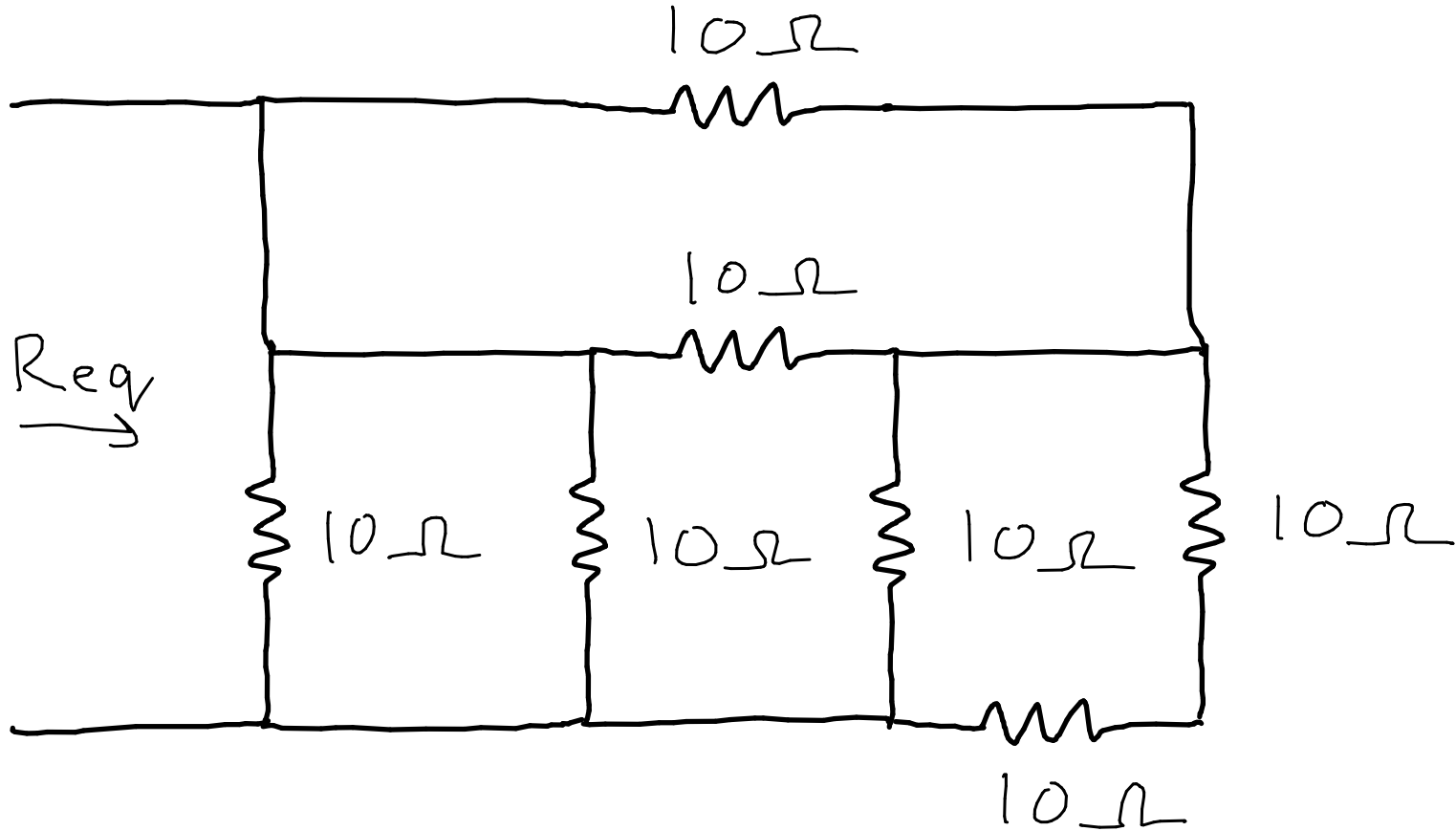
Ohm's law

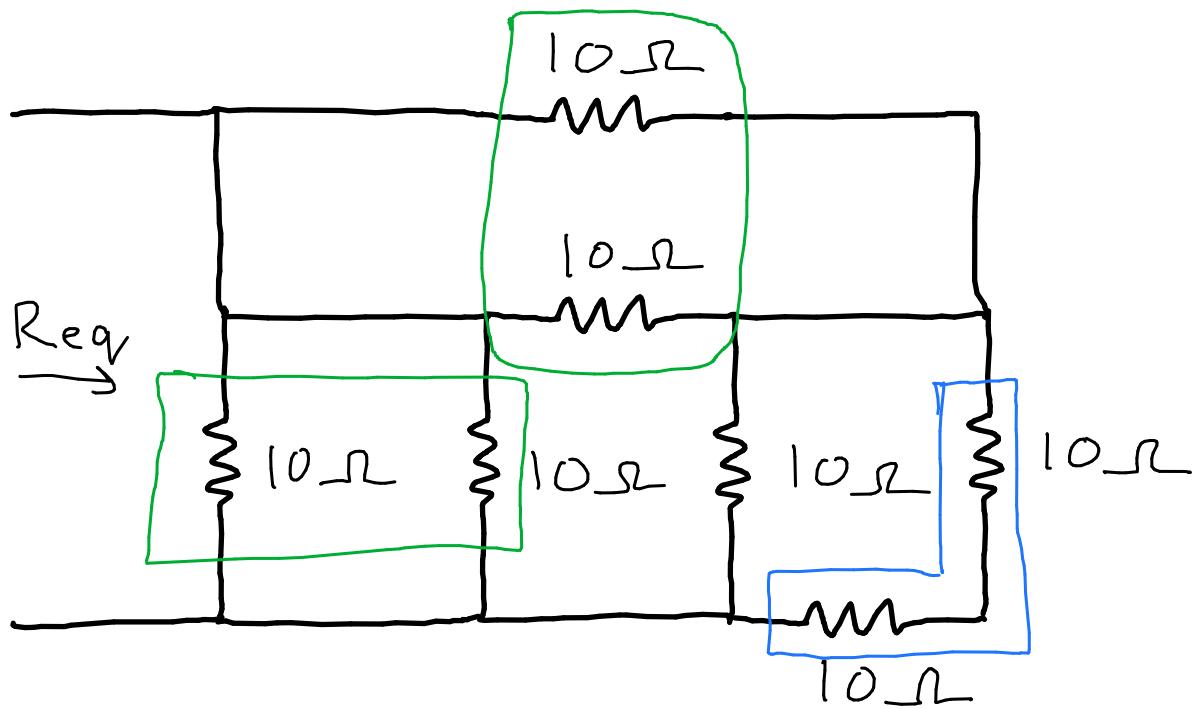
$$\begin{aligned} V_{20k} &= i_{20k} R_{20k} \\ &= 0.1 \times 20 \times 10^3 \\ &= 2000 \text{ V} \end{aligned}$$

Power law

$$\begin{aligned} P_{20k} &= i_{20k} V_{20k} \\ &= 0.1 \times 2000 = 200 \text{ W} \end{aligned}$$

2.29 Find R_{eq}





Parallel combinations

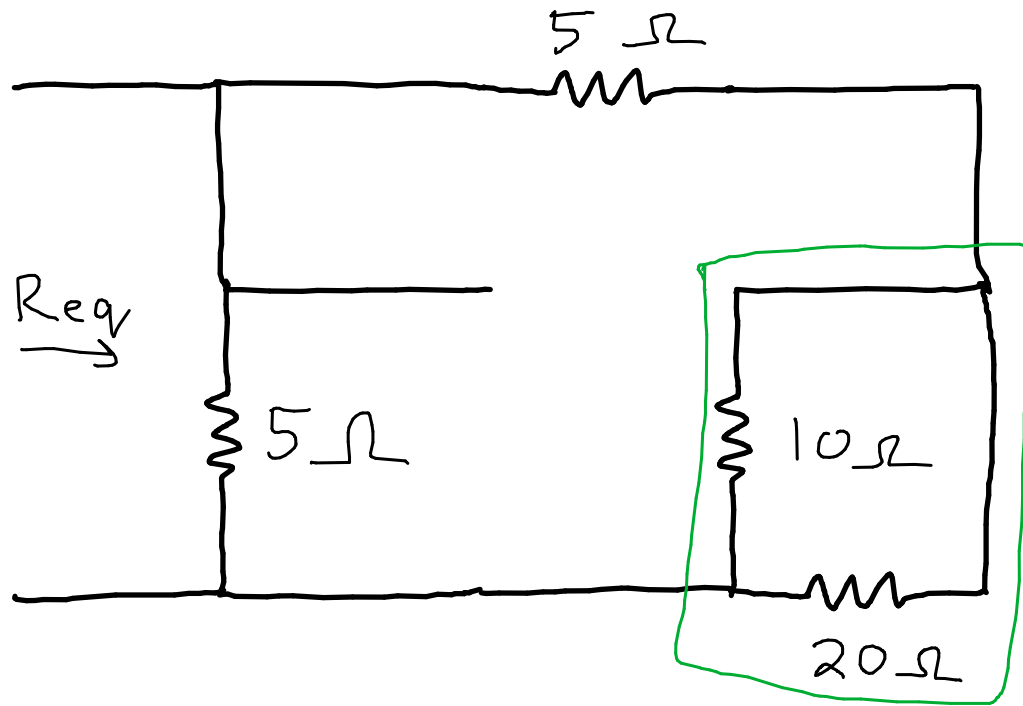
$$10 \parallel 10 = \left(\frac{1}{10} + \frac{1}{10} \right)^{-1}$$

$$= 5\Omega$$

series

combinations

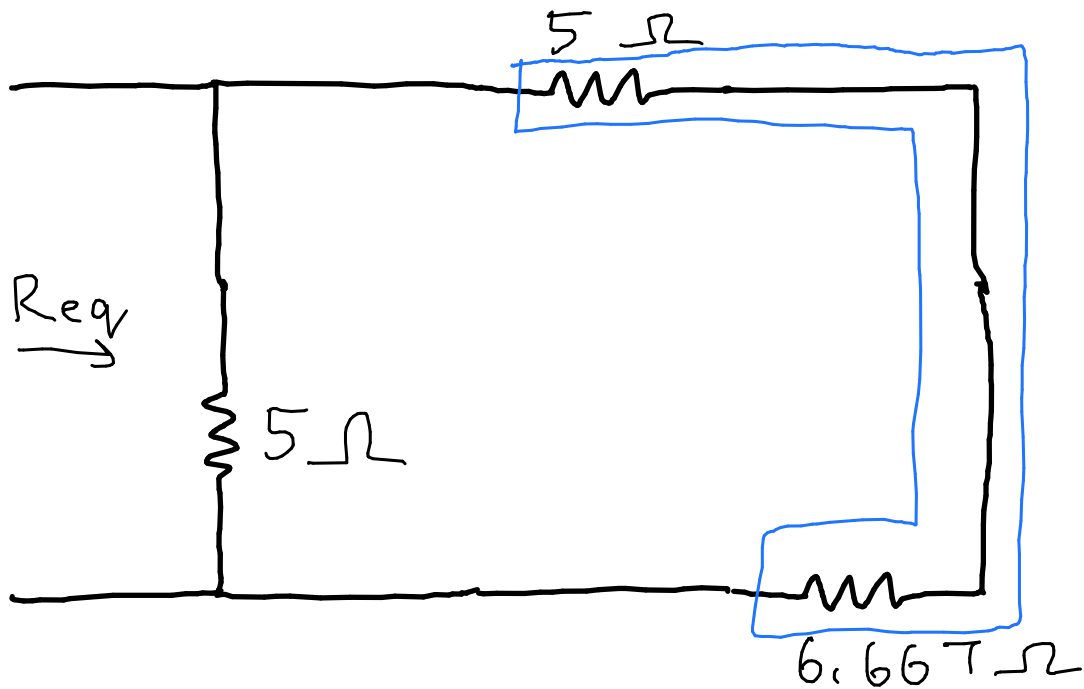
$$10 + 10 = 20\Omega$$



Parallel combination

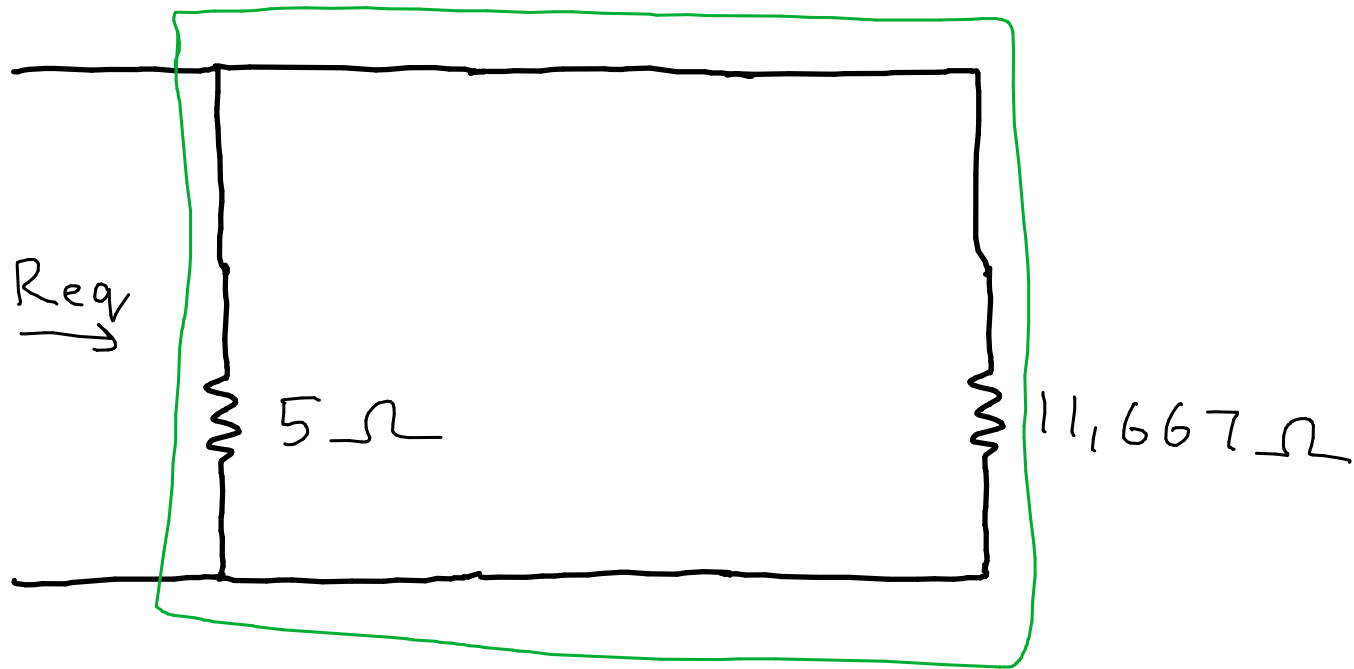
$$10 \parallel 20 = \left(\frac{1}{10} + \frac{1}{20} \right)^{-1}$$

$$= 6.667 \Omega$$



Series combination

$$5 + 6.667 = 11.667\ \Omega$$



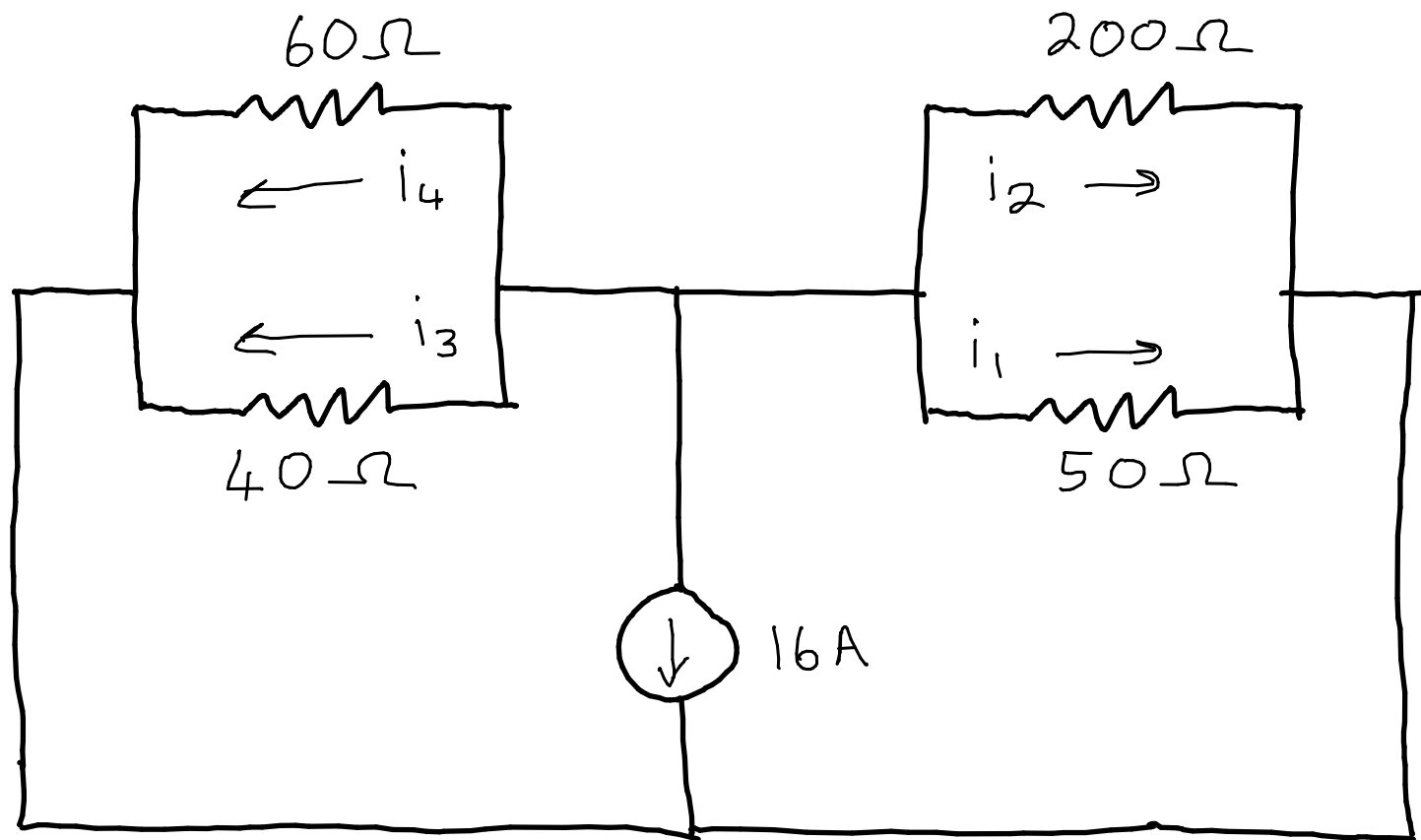
Parallel combination

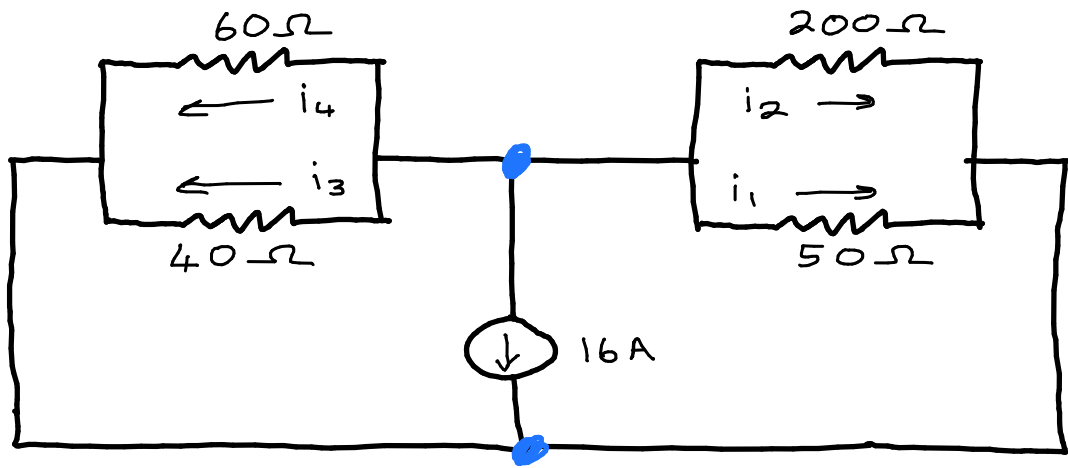
$$R_{eq} = 5 \parallel 11,667$$

$$= \left(\frac{1}{5} + \frac{1}{11,667} \right)^{-1}$$

$$= 3,5\ \Omega$$

2.32 Find i_1 through i_4





current division

$$i_n = \frac{i G_n}{G_{eq}}$$

KCL

$$i_1 + i_2 + i_3 + i_4 = -16A$$

$$G_{eq} = G_1 + G_2 + G_3 + G_4$$

$$= \frac{1}{50} + \frac{1}{200} + \frac{1}{40} + \frac{1}{60}$$

$$= \frac{1}{15}$$

$$i_1 = \frac{-16 \times 1/50}{1/15}$$

$$= -4.8 \text{ A}$$

$$i_2 = \frac{-16 \times 1/200}{1/15}$$

$$= -1.2 \text{ A}$$

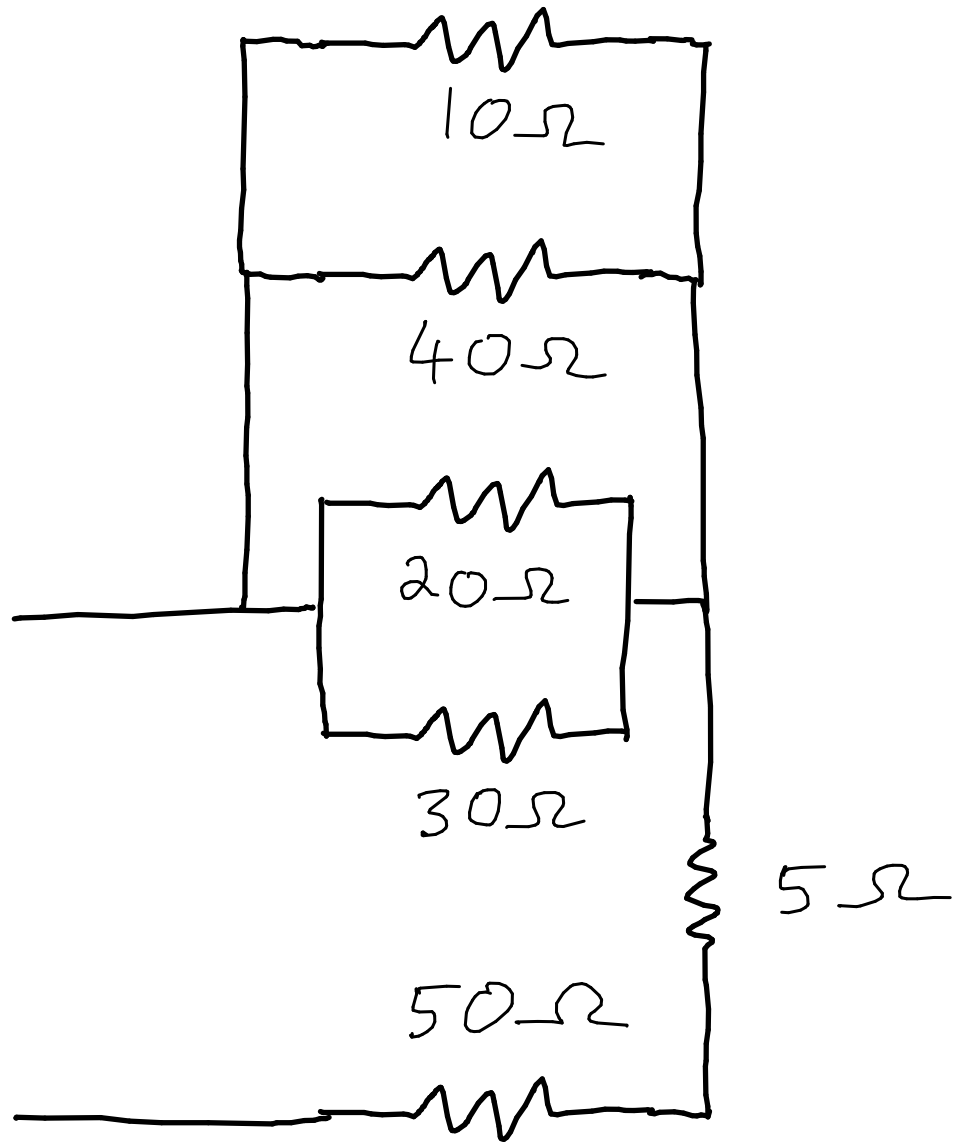
$$i_3 = \frac{-16 \times 1/40}{1/15}$$

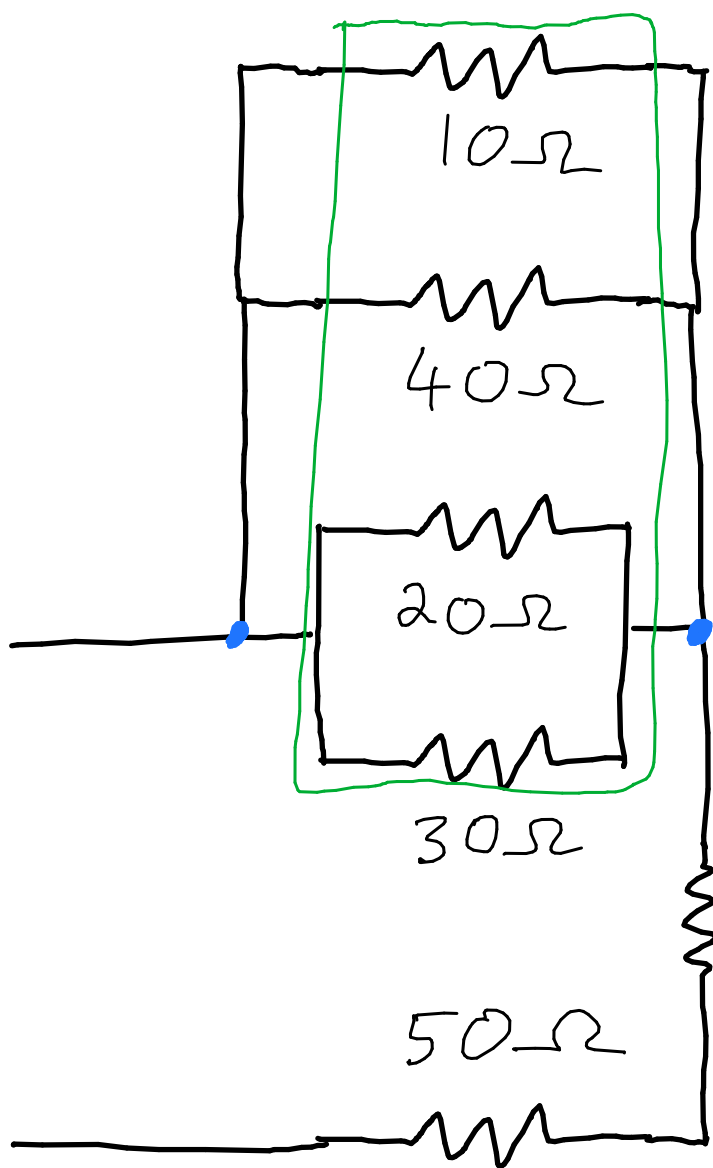
$$= -6 \text{ A}$$

$$i_4 = \frac{-16 \times 1/60}{1/15}$$

$$= -4 \text{ A}$$

2.45 a) Find the equivalent resistance





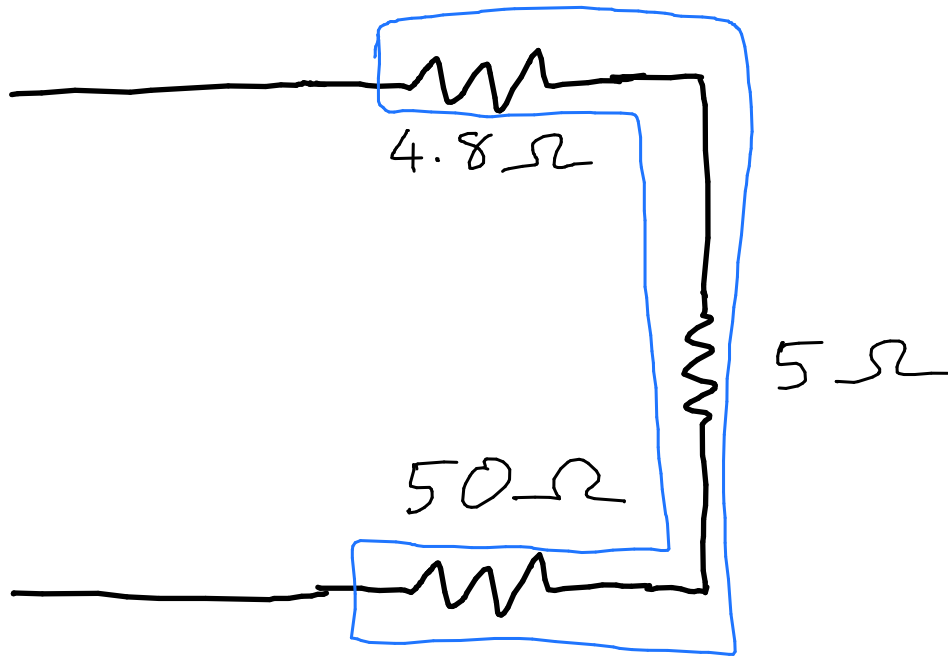
parallel combination

$$30 || 20 || 40 || 10$$

$$= \left(\frac{1}{30} + \frac{1}{20} + \frac{1}{40} + \frac{1}{10} \right)^{-1}$$

$$= 4.8\Omega$$

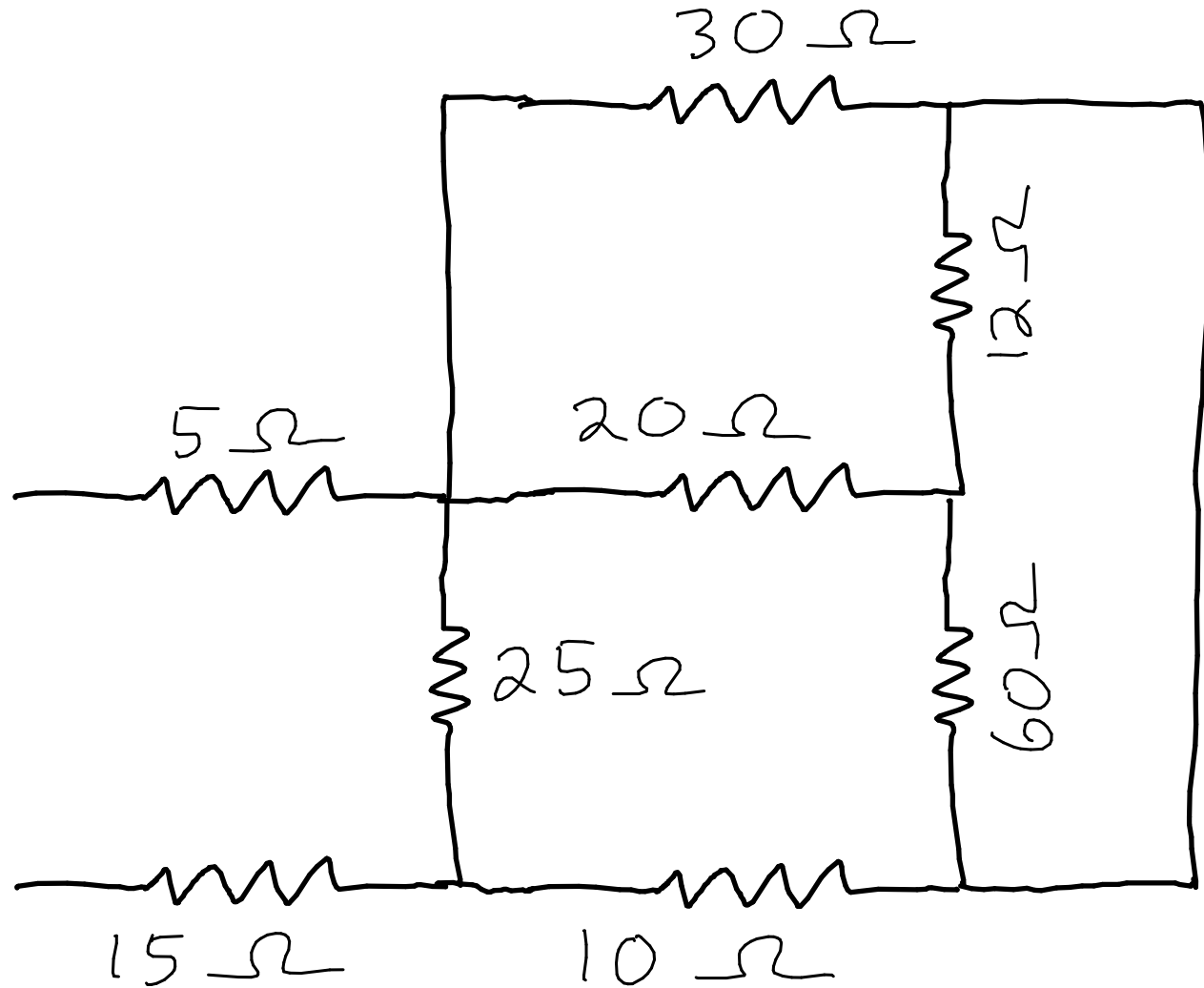
$$5\Omega$$

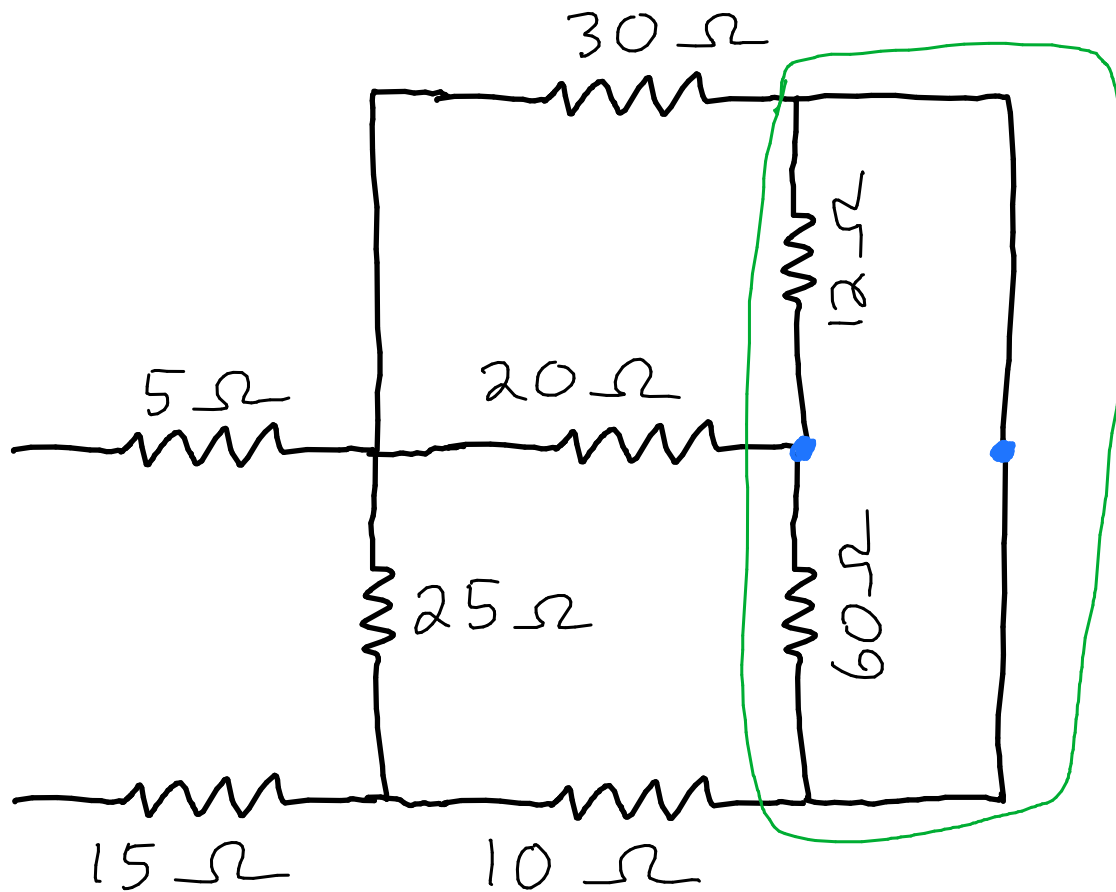


series combination

$$\begin{aligned} R_{eq} &= 4.8 + 5 + 50 \\ &= 59.8\ \Omega \end{aligned}$$

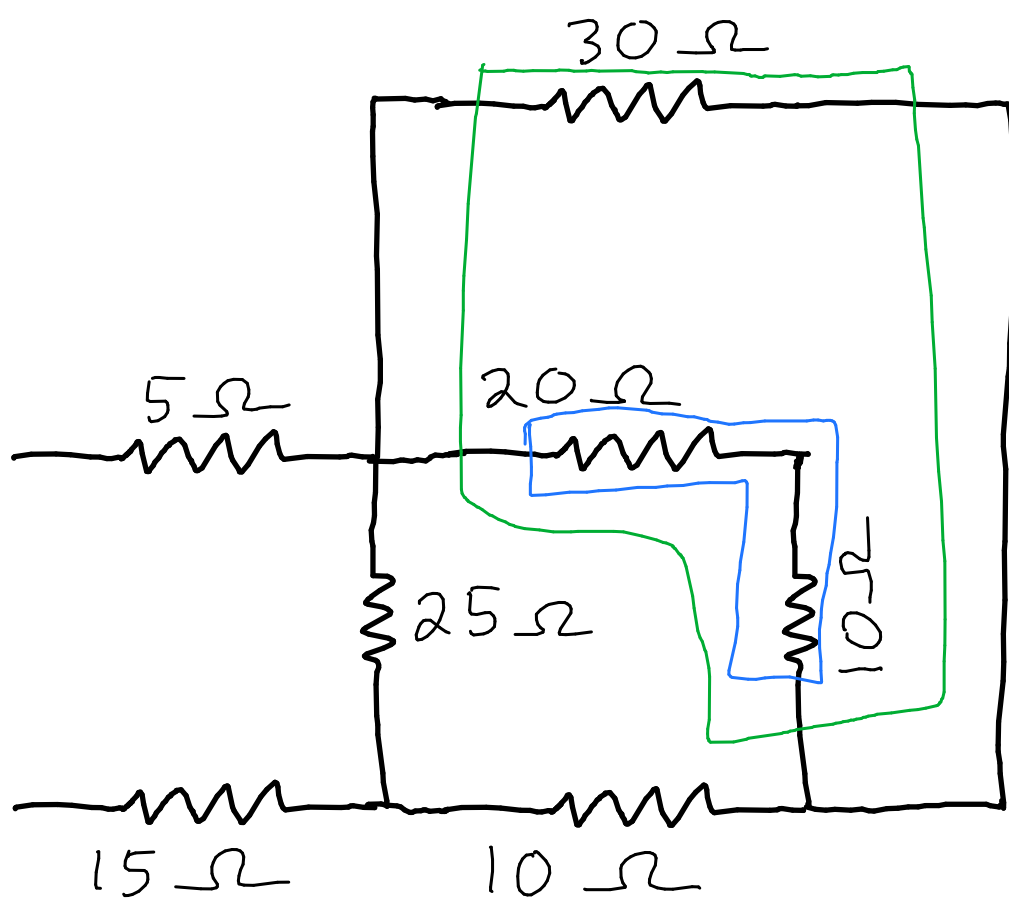
2.45 b) Find the equivalent resistance





parallel combination

$$12 \parallel 60 = \left(\frac{1}{12} + \frac{1}{60} \right)^{-1}$$
$$= 10\ \Omega$$

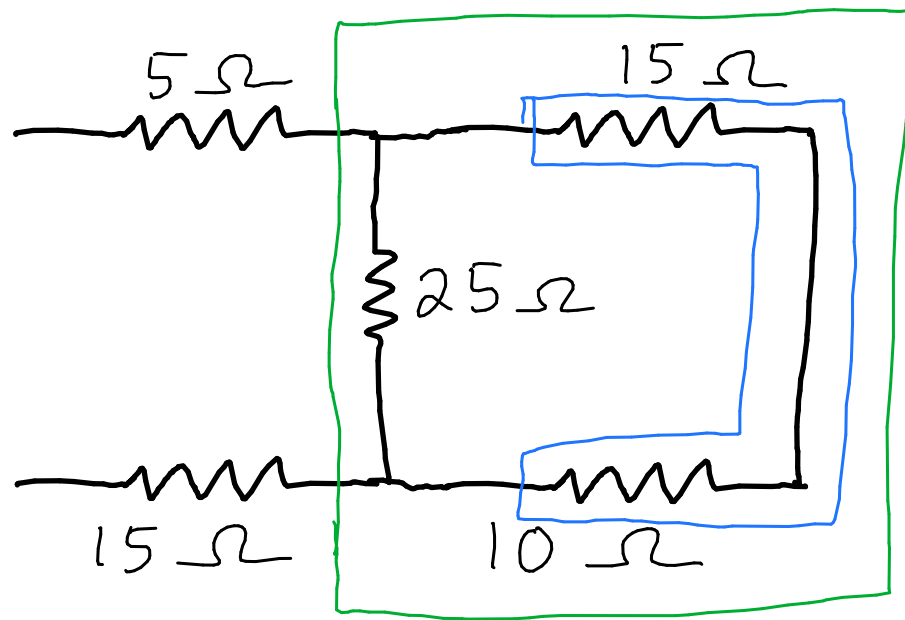


Series
combination
 $10 + 20 = 30\ \Omega$

parallel combination

$$30 \parallel 30 = \left(\frac{1}{30} + \frac{1}{30} \right)^{-1}$$

$$= 15\ \Omega$$



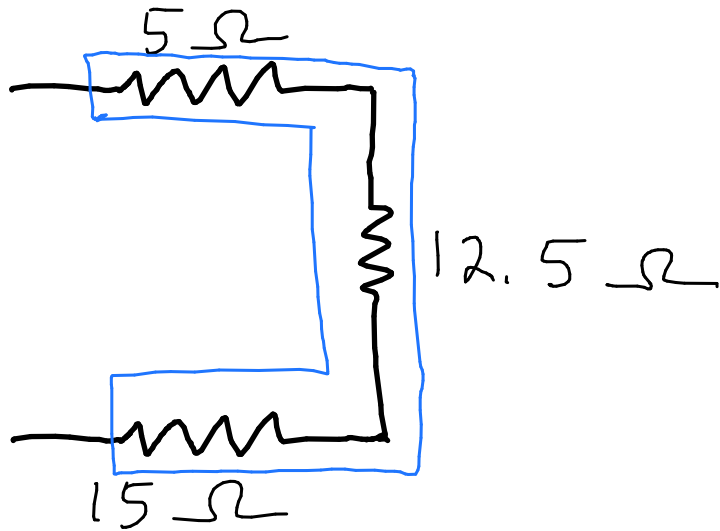
Series
Combination

$$15 + 10 = 25\ \Omega$$

parallel combination

$$25 || 25 = \left(\frac{1}{25} + \frac{1}{25} \right)^{-1}$$

$$= 12.5\ \Omega$$



Series combination

$$R_{eq} = 15 + 5 + 12.5$$

$$= 32.5\ \Omega$$